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High altitude exposure and the epidemiology of ischemic stroke, A literature review

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Abstract

Introduction

About 5.7% of the world population resides above 1,500 m. It has been hypothesized that acute exposure to high-altitude locations can increase stroke risk, while chronic hypoxia can reduce stroke-related mortality.

Objective

This review aims to provide an overview of the available evidence on risk prevalence, risk of stroke incidence, and or mortality when exposed to acute or chronic hypoxia.

Design

A Systematic literature review of the available information was performed to answer the link between hypobaric hypoxia due to high altitude exposure and stroke. The following libraries, repositories, and databases were accessed using the University of Southampton library tool Delphis: AMED, EMBASE, Cochrane Library, PubMed, MEDLINE, and Europe PMC. The Latin-American bibliographic database Scielo was also included in a separate search from its website repository.

Results

We reviewed a total of 1,518 abstracts retrieved during the first step of the literature review process. The authors included 191 manuscripts, while the rest of the abstracts were removed from the process for not meeting the inclusion criteria. Only 11 documents were included in the literature review, analysed in-depth, and discussed in the present document.

Conclusions

The risk of developing stroke seems to be higher in those living at low altitudes (< 2,000 meters), and the protective hypothesized effect seems to occur between 2,000 up to 3,500 meters of elevation. Increased irrigation due to angiogenesis and increased vascular perfusion might be the reason behind improved survival profiles among those living within this range. In contrast, above 3,500 meters, the presence of polycythaemia and other factors such as increased blood viscosity and the presence of a proposed hypercoagulable state might increase the risk of developing stroke among those acutely exposed to very high altitudes.

Keywords: Stroke; High altitude; Hypoxia; Thrombosis; angiogenesis; Review

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Strengths and limitations of this study

- This systematic review represents, to our knowledge, the first and only comprehensive examination of the intricated relationship between high altitude exposure and stroke available worldwide.
- The information provided within this review invites further researchers to explore the relationship between high-altitude related hypoxia, time of exposure, acclimatization and adaptation and the risk of developing stroke among highlanders.
- Future research is needed to obtain a definitive answer; nevertheless, the information provided in this document can be used as an updated guide of the possible role of high-altitude exposure as a risk factor for developing a stroke.
- The lack of follow-up data and the scarcity of information around stroke size, stroke severity or stroke sequela are important limitations when concluding the role of high altitude living and the risk of developing cerebrovascular events.
- This review only found observational studies, therefore causality cannot be inferred from this analysis before completing further research.

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Introduction

Stroke, also known as a cerebrovascular event, is a severe and often life-threatening medical condition that appears when the blood supply to part of the brain is interrupted [1]. It causes a rapid, acute, and potentially progressive brain function loss due to a disturbance in blood supply, oxygen, and nutrients that lasts at least 24 hours [2].

Cerebrovascular events or stroke is the second leading cause of death worldwide, affecting more than 16 million people each year[1]. Around 1 in 6 men and 1 in 5 women will have a stroke in their lifetime, and it is the third leading cause of disability worldwide [3,4]. Stroke affects people of all ages, though the causes of a stroke at a younger age are often different from those at older ages, especially in terms of risk factors and severity [5,6].

Clinical manifestations usually depend on the severity of the anoxia, the brain's location, the type and subtype of stroke, and the presence of other individual risk factors. The most common clinical manifestations are sudden unilateral weakness or numbness, diplopia, ataxia, and aphasia [1].

Stroke risk factors are often classified as traditional and non-traditional, and from this, they can be modifiable or non-modifiable [7]. In general terms, these risk factors have been classified as those which cause an ischemic stroke or haemorrhagic, being the most common and recognized as displayed in table 1.

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Table 1 Traditional modifiable and non-modifiable risk factors of stroke

Traditional risk factors			
		Non-modifiable	**Modifiable**
		Age	
		Sex	
		Ethnicity	
		Socio-economic status	
		Family History	
			Waist circumference
	Thrombotic	Arterial dissection	Alcohol misuse
			Obesity (BMI, WC, WHR)
			Diabetes
			Cigarette Smoking
		Atrial fibrillation	
Ischemic Stroke		Intracardiac thrombus	
	Embolic	Heart valve disease	Physical inactivity
		Trauma and fractures	Apolipoprotein B to A1
		Some types of Surgeries	Hyperlipidaemia***
	Systemic	Post-traumatic hypovolemia	
		acute systemic hypoxia	
		Vascular Malformations	Hypertension
	***ICH**	bleeding diatheses	Cigarette smoking
		Trauma	Obesity
			Waist -to-hip ratio
Haemorrhagic stroke		Cocaine Abuse	Diet
	****SAH**	Amphetamines	

*Intracerebral haemorrhage ** Subarachnoid haemorrhage *** High LDL serum levels

The risk of developing stroke increases with the presence of causal factors, which include arterial hypertension, atrial fibrillation, cigarette smoking, hyperlipidaemia, and diabetes mellitus [1]. Other modifiable factors are obesity, chronic kidney disease, excessive alcohol and cocaine consumption, sedentarism, psychological stress or depression [8–10]. Other factors such as Vitamin D or high-altitude may play a role but have been less studied than the traditional vascular risk factors [11–15].

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Rationale for high altitude exposure and stroke

Globally, at least 5.7% of the population lives above 1,500 meters, nevertheless, the association between high-altitude exposure and stroke has been poorly studied [12,13,16]. The available literature is based mainly on case reports due to acute exposure to extreme altitudes, rather than longer-term living among high-altitude dwellers [12,17–19]. The few reports available suggest that living in high-altitude regions (>2,500 m) increases the risk of developing thrombosis through hypoxia driven polycythaemia which leads to a hypercoagulation unbalance, that have been associated with increased risk for the development of atherothrombotic stroke [20–22].

Although most of the information comes from a case or case-series reports, no cohorts' studies have been published in this area. The only available information came from very few cross-sectional analysis that has found a significant association between living in high-altitude regions and having a greater risk of developing stroke, especially among younger populations [12,13,18,19].

Objective

To further explore this relationship, we have conducted a literature review of the available information in terms of the link between high-altitude chronic exposure and the risk of developing ischemic stroke.

Methods

Research Question

Does living at high-altitude increases the risk of developing ischemic stroke?

Study Design

To answer the link between hypobaric hypoxia due to high altitude exposure and stroke; a literature review of the available information was performed. The first step prior the analysis was conducting a review of the existing published evidence. To conduct a comprehensive study of the available literature we performed a systematic search within the main bibliographic databases (see below). The review search was conducted

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in accordance with preferred reporting for systematic reviews and meta-analyses (PRISMA) guidelines. No protocol was registered in PROSPERO for this review because of its small size.

Information sources

The literature review process was performed in English to cover the largest number of scientific databases and repositories containing academic literature from all over the world that are accessible for most researchers. The following libraries, repositories, and databases were accessed using the University of Southampton library tool Delphis: AMED, EMBASE, Cochrane Library, PubMed, MEDLINE, and Europe PMC. The Latin-American bibliographic database Scielo was also included in a separate search from its website repository.

Search strategies

We identified the specific population using the PICO question (population, intervention, comparator, outcome) In this review, we stipulated the question as every study available from 1960 to 2019 that include stroke and chronic high altitude exposure excluding case reports. The following terms were used during the search strategy within the tittle (TI) or the abstract (AB): High altitude OR High altitude Exposure OR High Altitude living OR High altitude dwellers AND combined with the complete or trunked MESH terms stroke OR Cerebrovascular accident OR CVA OR Ischemia OR Thrombosis OR Hemorrhag* OR Haemorrhag* OR subarachnoid haemorrhage OR subarachnoid hemorrhage OR haemorrhagic stroke OR hemorrhagic stroke OR Ischemic stroke.

Inclusion and Exclusion criteria

This literature review excluded all in vivo and in vitro studies, and all the data analysed that did not concern human studies. If the search term "stroke" referred to cardiac output or stroke volume, the results were also excluded. Case reports were excluded from the study. The data extraction length was set from 1960 to 2019, and any study that matched the search strategy and did not counterpose the exclusion criteria was revised. We set the cut-off point of short or long-term exposure in 28 days based on previously published data.

Bias assessment

In order to reduce risk of any type of bias, the data extraction process was performed by EO and SC independently and in different times. in order to minimize errors while gathering information from any primary study, any disagreements was resolved after reaching consensus. In addition, while assessing the quality of primary studies, the CASP critical appraisal tool was used to reduce confirmation bias.

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Ethical consideration

This secondary data analysis of publicly available anonymized data received ethical approval from the University of Southampton with the Faculty of Medicine Ethics Committee ERGO 51422.R3 number.

Patient and Public Involvement

No patient involved

Results

Literature review

Several anecdotal case reports and case series of people suffering from stroke-related disorders at high-altitude have been published since the late 1800s [23]. Despite this apparent relationship, very few well-conducted epidemiological studies have been conducted worldwide, and we have only included those studies that fulfilled our inclusion and exclusion criteria [12,13,18,19,24,25].

We included 1,518 manuscripts during the first step of the literature review process, and only those that fulfil the inclusion criteria were revised. After this process, the authors evaluated 191 abstracts. In the end, we included 11 manuscripts within this literature review.

Figure 1 Prisma chart diagram for reporting literature review search and metanalysis

Specific studies

Study 1: The first study published by Razdan et al. in 1988 looked into the prevalence of stroke in elevations above 1,500 m [26]. They found that the crude prevalence of stroke was higher than the populations residing below this elevation (Table 2) [26].

Study 2: In 1990, Annoml et al. published an article about the incidence of cerebrovascular accidents in children with sickle cell disease residing at high and low altitudes in Saudi Arabia. They found that 4% and 5% of the 400 children included in the study who were living at low and high altitudes developed stroke.

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The authors did not explore any causal relationship or controlled for confounders. The results presented in that study are not significant, considering the total number (n=8) of children with sickle cell diseases and stroke [27].

Study 3: In 1994, Garcia-J reported the prevalence of risk factors of developing stroke at 2,500 m in a community in Peru; nevertheless, the study focused on the association between risk factors and stroke and not in the relationship with elevation[28].

Study 4: In 1995, Jaillard et al. published one of the most cited stroke documents at high-altitude. A document from Perú described the results from a "door to door" survey in Cuzco, a city located at 3,380 meters above sea level. They reported a crude prevalence rate of 6.47 per 1,000 population; nevertheless, comparisons were performed using standard populations from other locations, while not giving enough information about the association of high-altitude and stroke per se. This study's evidence is limited since the comparison was performed with other reports at lower elevations; therefore, the conclusions should be taken cautiously [18].

Study 5: A Population-based analysis published by Jha et al. in 2002 reported a significantly higher incidence of stroke among patients admitted to a single tertiary hospital in India. They classified those patients that came from lower altitudes (<3,000 m) and higher altitudes (>3,000 m). They reported that the majority of stroke cases at high-altitude were presented in otherwise-healthy young men (<45 years old) exposed to high-altitude for months (nine months on average). The hospital admission rate at high-altitude was reported to be 12 times greater (12.8/1,000) than at lower altitudes (1.0/1,000), being this results statistically significant [12].

Study 6 In 2003, Niaz et al. conducted a case-control study to describe the prevalence of stroke above 4,571 m. They included 4,000 soldiers (20-40 y/o) from a military hospital to observe stroke occurrence. They reported 10 cases of stroke among young men patients from the high-altitude group versus only one case from the lower land cohort. Although no further information about size effect or standardization of the sample was provided, the researchers concluded that living at elevations above 4,500 m was associated with ten times more risk of developing stroke. The authors concluded that chronic (> 28 days) hypobaric hypoxia exposure it is associated with a higher risk of developing stroke, being massive ischemic stroke the most frequent presentation [13].

Study 7 In 2004, Mahajan et al. conducted a cross-sectional study in Himachal, Pradesh, India, among 100 patients hospitalized in a tertiary level hospital located at an altitude of 2200 meters. They reported that the incidence of stroke was lower when compared with previously published studies from India. Although information from the type of stroke and clinical features are available, the role of elevation was not discussed in depth. The authors compared the results with other studies, but no confounding analysis was made for the current report [29].

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Study 8: An ecological study published by Faeh et al. in 2009 reported that stroke mortality is lower in higher locations in Switzerland (Faeh D et al., 2009). They reported that mortality from stroke decrease 12% per every 1000 m gained in elevation. The effect of altitude on the cardiovascular and cerebrovascular disease was assessed using sociodemographic information, places of birth and residence among men and women between 40 and 84 years of age living at altitudes of 259 to 1960 m. The protective effect of living at higher altitudes on coronary heart disease and stroke mortality was consistent and became stronger after adjustment for potential confounders [19].

Study 9: A study presented by Ezzati et al. in 2011, used from the National Elevation Dataset, National Centre for Health Statistics, and US Census. The authors analyzed the crude association of mean county altitude with life expectancy and mortality from ischaemic heart disease (IHD), stroke, chronic obstructive pulmonary disease (COPD), and cancers. Living at higher altitudes showed a beneficial association with ischemic heart disease mortality, but the results were not statistically significant for stroke [30].

Study 10: A study published by Dhiman et al. 2018 reported the evolving Pattern and Outcome of Stroke at Moderate Altitude. This prospective, observational study was carried at 2,000 meters above sea level and included the clinical features and risk factors profile of 235 patients diagnosed with a stroke. The results only compared how the data compared with a report published 15 years ago, and the authors concluded that the occurrence of stroke decreased among hospitalized patients compared to the 2004 report [29,31].

Study 11: An Ecological study design published by Khattar et al. in 2019 demonstrated that ischemic stroke is most likely to occur at elevations above 3,000 m and although this diagnosis is not technically considered stroke, the physiopathology of it might indicate some relationship with the development of ischemic cerebrovascular events. Despite some of these findings that targeted high-altitude as a possible risk factor for developing stroke, in 2009 [32] (Table 2)

Table 2 Critical appraisal and summary of the literature.

Author and year	Tittle	Sample, setting and elevation	Aim	Design, type and length of Exposure	Main findings	Strengths	Limitations and confounding. Bias Analysis

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Razdan et al. 1988	Cerebro-vascular Disease in Rural Kashmir, India	-Kuthar Valley, India -Age (5 and 65) years. -80% of pop. Are shepherds and cultivators while others are immigrants. -Elevation 1,530 m.	To find the prevalence and trends related to stroke in this region	-A cross sectional survey to find if population has suffered from stroke. -First part of the survey was delivered by trained med student. -Second part included two neurologist who visited the houses of those who were likely to have suffered from stroke. - An ex post-facto relationship was performed using the Garraway diagnosis criteria -Long term hypoxia exposure was identified. The final outcome was stroke prevalence. No confounder factors were controlled, and the presence of multi-ethnic immigrants might have played a role in the interpretation.	- 91 cases of stroke were detected - Crude prevalence rate 143/100,000 - Proportion in men (69,23%) and women (30,77%) - >40 y/o 630/100,000 ->15 y/o 244/100,000 - the most common risk factor among the elderly was hypertension - In young stroke patients valvular heart disease and postpartum cerebral venous thrombosis.	-First of its kind in India and probably the world. In exploring the role of high-altitude. -A team of neurologists interviewed and examined all suspected cases of strokes including 500 home visits. -A good sense of case control matches was reported	-Elevation not greater than 2,000 meters, hypobaric hypoxia not enough to cause a physiological long term-response -They did not differentiate between cerebral infarction or haemorrhage -They did not consider smoking habits
Annoml et.al 1990	Cerebrovascular accidents in children with sickle cell disease residing at high and low altitudes of Saudi Arabia	-Patients with sickle cell disease residing at high and low altitudes -3,000 m	- To compare the clinical and radiological indications of stroke in sickle cell patients constantly residing at low and high-altitudes.	-Analyses of 8 cases of stroke among children with sickle cell disease -extracted from 400 cases detected in Saudi Arabia. -Sample size was not representative and the differences between low and high altitude was only one patient. -The risk of interpretation bias is present, and we recognized this limitation.	-The incidence of stroke in children with sickle cell disease similar regardless elevation. -Clinical manifestations were more severe in those residing al low altitude.	Complete studies were performed on children such as physical exams, blood tests and imaging studies, thus, diagnosis was accurate.	-The number of subjects is too small to extrapolate results with confidence. -No confounders were controlled, and the difference found in both elevations was not significant neither clinically relevant.

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Garcia, J 1994	Cerebrovascular Disease: Risk Factors analysis of Natives living at high-altitude.	-50 patients with a confirmed diagnosis of stroke and 52 healthy patients -2,500 m	To understand the role of different risk factors in patients located at high altitudes	Retrospective case-control study - Population selection based on criteria (cases), random population (controls) - Survey method and information on risk factors was obtained through investigation sheets - The factors were divided into 2 groups: causal and aggravating - Diagnosis of stroke was corroborated by neurologists in all cases.	- Atherosclerosis was the most important risk factor for CVD in the highland natives. - Hypertension and nonrheumatic atrial fibrillation came afterward. - Old age, frequent alcohol consumption and prior CVD were contributor factors. - Cerebral infarction was the most frequent type of CVD, it was present in 90% of cases. -The main clinical category of cerebral infarction was atherothrombotic in 56% of cases, with a notable predominance in the male sex.	- The study analyses the relationship between hypertension with developing of stroke, in addition to other triggers of this disease. - Diagnosis of stroke was corroborated by neurologists in all cases which decrease bias - This study determines that the prevention of risk factors is the key for the prevention of this disease	The sample used in the study is small -The important risk factors that were not included in this study, are ischemic heart disease and diabetes, are clinically proven that these factors are also causes of developing stroke
Jaillard, 1995	Prevalence of Stroke at High-altitude (3380 m) in Cuzco, a Town of Peru A Population-Based Study	-3,246 subjects interviewed -men and women older than 15 years old were included -97% Mestizos, 1.3% white, 1.1% Quechua, 0.1% Aymara, and 0.2% were of other racial groups -3,800 m	to study the prevalence of stroke and its association with stroke risk factors in a high-altitude location.	A door to door population-based analysis using a survey as a tool for collecting data.	-Age, polycythemia, high consumption of alcohol, and area of residence were associated with higher stroke prevalence. -Sedentarism and high income was associated with higher OR of stroke in El Cuzco - 21 strokes cases were identified	Clinical evaluation of the patients with stroke was performed by two neurologists, offering an accurate diagnosis *ex post facto*.	-No control group at lower altitudes. -No method of randomization was mention. the diagnosis was established on clinical criteria, and no attempt was made to distinguish haemorrhagic from ischemic strokes.

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Jha et. al 2002	Stroke at High-altitude: Indian Experience	-Clinical evaluation of 2,184 patients transferred from different medical units situated at different elevations and managed at the main Hospital in Chandimandir, India. -Soldiers (22 to 48 y/o) -Clinical reports of 30 cases of stroke at high-altitude -4,270 m	-To determine the relationship between stroke and high-altitude.	-Information was prospectively collected from the Command Hospital in India -A detailed neurological and systemic examination was carried out for all the cases.	-Long-term stay at high-altitude was associated with higher risk of developing stroke. -Ischemic stroke was the most common among the type of strokes. -22 cases of IS, 2 ICH, 4 TIA and 2 had CVT - Out of 30 cases, 28 were of stroke in young, categorized as every case before the age of 45. - Polycythaemia with Hb ranging from 16.2 to 22 g/dL21 was seen in 21 of the 28 reported cases representing more than 75%. -Protein C and S deficiency was found in 1 case in each of the groups. - CT scan showed massive infarcts involving at least 50% of one cerebral hemisphere in 12 cases, demonstrating more severity. -Multiple infarcts were seen in one case.	-A detailed neurological and systemic examination was carried out. -Cases were investigated with blood counts, lipid profile, cardiac evaluation, and CT scan/MRI. -Coagulation parameters were studied in some cases.	-They did not test for factor V or the Leiden coagulation factor. -They could not conclude that familial thrombophilia played a role in these high-altitude events, although is unlike this occur in this type of population.

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Niaz et al. 2003	Stroke at high altitude: Indian experience	-The study was performed in Chandimandir, India -Sample was based on 30 men soldiers - Age ranged from 22 to 48 years. -Elevation: 4,200 m.	-To identify the presence of stroke using different clinical, imaging and laboratory assessment techniques to determine the presence or not of CVA.	-A cohort analysis of stroke prevalence and mortality from 1998-2000. - A detailed neurological and systemic examination was carried out. Cases were investigated with blood counts, lipid profile, cardiac evaluation, and CT scan/MRI. Coagulation parameters were studied in some cases.	-Stroke hospital admission rates was higher among high altitude residents (13.7/1000) versus low altitude (1.05/1000) -The mean high-altitude stay was 10.2 months -Age ranged from 22 to 48 years (mean 33.4 yr). -Soldiers had no pre-existing risk factors except for smoking (in four cases). -22 cases were ischemic stroke, 2 intracerebral haemorrhages and 4 were reported as TIA/RIND (transient ischemic attack/reversible ischemic neurological deficit). The authors reported 2 cases of cerebral venous thrombosis.	-This study is important since included otherwise healthy young soldiers and finding stroke among this group of people is not common. - Elevation was considerable high and the detailed neurological and systemic examination was carried out offered complete information about sociodemographic and clinically relevant data.	-The fact that not all the soldiers were given all the laboratory tests may have conferred some limitation on the work
Mahajan et. al 2004	Stroke at moderate altitude	-this study was performed in Himachal, Pradesh in the Sub-Himalayan ranges. -100 patients admitted to the hospital -age 20 to 85 years, -elevation: 2,200 m	To understand the clinical profile, presence of various risk factors for stroke at moderate altitude.	-A cross sectional analysis of stroke admission in a tertiary hospital from India	-Patients aged up to 45 years were defined as stroke in young. -Complete clinical, radiological and neurological examination was performed.	-This is the first study performed at a moderate altitude of 2,200 meters in India. -A complete analysis of stroke clinical features is presented	-No comments about the biological plausibility of the protective effect of altitude was made in the documents. -No control group was used to asses the differences with other populations. -The results were compared with other previously published report and no controlling was evidenced while using those reports.

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Faeh et.al 2009	Lower Mortality from Coronary Heart Disease and Stroke at Higher Altitudes in Switzerland.	Men and women between 40 and 84 years. -259 to 1960 m.	Examine mortality from coronary heart disease (CHD) and stroke in its association with high-altitude living.	-A longitudinal analysis of mortality data from 1990 to 2000. Sociodemographic information, places of birth, residence and previously mobilization was obtained from the Swiss National Census-based record linkage study. - Data was obtained from the Swiss National Cohort, a longitudinal, census-based record linkage study.	-Living at higher altitude was associated with less CHD and less stroke mortality. -Mortality from CHD significantly decreased with increasing altitude as well as stroke (-12% per every 1,000 m	-First longitudinal study with individual data of a large homogeneous and representative population showing a protective effect of altitude on CVD mortality. - Consistent with results, other studies showed that the decrease in mortality with increasing altitude was stronger for CHD than for stroke.	The clinical risk factors of classic behaviour were not adjusted. -The variable for urbanity was artificial - The study not able to adjust for classic behavioural and clinical risk factors - The analysis cannot define causal relationships with risk factors or prove causal pathways for the effect of altitude.
Ezzati et.al 2011	Altitude, life expectancy and mortality from ischaemic heart disease, stroke, COPD and cancers: national population-based analysis of US counties	-Mortality and population data for 2001 through 2005. -12.1 million death records -Data from the National Centre of Health Statistics in US counties -500-1000 m -1000- 1500 m -1500 > m	Inspect the relationship of life expectancy and mortality from particular diseases in relationship to altitude.	-Ecological study of the association of mean county altitude with life expectancy and mortality from IHD, Stroke, COPD and cancer. -Results were adjusted for sociodemographic factors, migration, solar radiation, cumulative exposure to smoking and altitude.	- Counties above 1500 m had longer life expectances than those within 100m of sea level by 1.2 - 3.6 years for men and 0.5 -2.5 years for women. - Living at higher altitude may have a protective effect on IHD and a harmful effect on COPD. - The association between altitude and life expectancy for stroke and cancers were no statistically significant	- Is the first study to examine the relationship of altitude with life expectancy and with mortality from leading chronic diseases using consistent and comparable data and methods. - The large number of deaths in the vital registration data led to robust estimates of deaths rates and life expectancy.	- Data on how population and deaths are distributed by elevation within each county was not provided. -The lack of information about habits and risk factors might jeopardize the conclusions since people at high might be otherwise healthier even though high altitude maybe a risk.

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Dhiman et. al 2018	The Evolving Pattern and Outcome of Stroke at Moderate Altitude	-Men and women with an average age of 62 years Sub-Himalayan region of India. -235 patients with stroke were consecutively admitted into a tertiary hospital. -2,000 m	The aim was to compare the clinical profile, risk factors, and outcome in hospitalized patients of stroke in a Tertiary Care Hospital situated at moderate altitude	-A prospectively collected Study in India. -A comparative analysis was performed with a previous study performed 15 years prior of the current study.	-Ischemic stroke was noted in 74%, and 26% had haemorrhagic stroke (HS). -Male accounted 58% of the cases and female for 42%. -Overall HS had poorer outcome. -There occurrence of stroke has decreased among hospitalized patients at moderate altitude.	The results of the present study were compared with the study conducted 15 years ago therefore a before and after analysis was possible.	-Elevation was assessed but no low altitude controls were presented or used. -The control group was from the same elevation, but no description of prior methodology was provided.
Khattar et.al 2019	Cerebral Venous Thrombosis at High-altitude: A Retrospective Cohort of Twenty-one Consecutive Patients	-The study cohort comprised 21 patients in Nepal. CVT was the diagnosis of interest. -elevation 3,048 m	Investigate the characteristics and treatment outcomes of patients who suffered CVT at high-altitude in eastern Nepal.	- Retrospectively reviewed all patients presenting with clinical and radiographic evidence of cerebral venous sinus thrombosis treated. - Neuroimaging evaluation and preoperative planning: Brain computed tomography (CT) was obtained in every patient prior to a definitive diagnosis.	21 patients of which 76% were men with an average age of 56. -men were found to have a higher risk for CVT at high-altitude -All patients presented with evidence of haemorrhagic conversion on the initial brain CT.	-A complete set of clinical data and outcomes were analysed.	-small sample size make it difficult to extrapolate the results -No controls at lower altitudes were used.

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Discussion

The results from this review suggest that stroke seems to be more likely to occur in very high-altitude locations (>3,500 m) when the exposure is longer than 28 days, especially among younger people (< 45 y/o). On the other hand, when people live above 1,500 m and below 3,500 seems like there is a protective effect for stroke, probably triggered by better adaptation to hypoxia, efficient enough to reduce the likelihood of dying when compared to lower altitudes; nevertheless, no information is available about the exact point in which this protective effect become a risk factor.

It has been challenging to define how high altitude exposure can be defined and where the threshold is located in terms of mild or severe hypoxia [33]. For instance, Imray in 2011 uses a classification of high-altitude exposure according to the recommendations from the International Society of Mountain Medicine, a categorization that seems to be the most pragmatic [34]. The author defined low altitude everything located below 1,500 m, moderate or intermediate altitude between 1,500 to 2,500 m, high-altitude from 2,500 to 3,500 m, the very high-altitude from 3,500 m to 5,800, more than 5,800 extreme high-altitudes and above the 8,000 m is considered the death zone [34].

As humans acclimatize to high-altitude, adverse and often mild secondary effects can occur in response to hypoxia. Some of these adverse effects are linked to the increased blood viscosity due to polycythemia, augmented pulmonary arterial pressure, and sometimes, they are linked to a proposed hypercoagulation unbalance [20,22].

These consequences might be increasing the risk of forming an atherothrombotic plaque resulting in a stroke or myocardial infarction (MI) or venous-thrombotic events, resulting in deep venous thrombosis (DVT) or pulmonary embolism (PE) [35–37]. Although information about the time of exposure is scarce, the longer the exposure, the higher the risk [19,38].

Acute exposure to hypobaric hypoxia produces several compensatory physiological effects that can last hours, days, months, or years. The essential mechanisms are: increasing the heart and respiratory rates, a secondary polycythemia, haemoconcentration derived from reduced plasma volume caused by respiratory evaporative water loss and polyuria, and increased ventilatory response [33,39,40]. When acute exposure last longer than 28 days, more efficient and prolonged mechanisms take place, including sustained polycythemia, endothelium changes, reduced vascular resistance, nitric oxide-mediated hypotension, and angiogenesis [41–44]. Acute exposure to high-altitude hypoxia triggers a series of events that produce a hypercoagulable state [22]. This hypercoagulability state is boosted by dehydration, haemoconcentration, and polycythemia. When combined with dehydration (due to tachyphemia and extenuate physical activity) and limited mobilization (sleeping in tents and secluded spaces), these factors produce the perfect scenario for increasing vascular stasis and thrombosis [20,33,45].

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When humans are exposed continuously to hypoxia, they develop adaptative mechanisms that are far more efficient than those observed in newcomers [46–49]. These last-longing mechanisms include anatomical (wider chests, shorter and lighter bodies, etc.), embryological (smaller fetus and placentas), circulatory (improved maximum flow output and higher pulmonary arterial pressure), and respiratory adaptations (improved hypoxic ventilatory response and oxygen diffusion capacities) [48,50–52]. Chronic exposure to hypobaric hypoxia leads to the development of more subtle compensatory mechanisms. These factors include long-term erythrocytosis, angiogenesis, capillary remodelling, and an improved ventilatory response [53–56] (Figure 2).

Once the general context of acute or chronic hypobaric hypoxia has been described, the main intrigue is which elevation is enough to generate compensatory mechanisms capable of reducing the risk of developing stroke and when these mechanisms become detrimental. After reviewing the current literature, the information available suggests that a window around 2,000 to 3,500 meters of elevation might be enough to generate some protective mechanisms (i.e., angiogenesis or vascular remodelling) against stroke [19,41,44].

In elevations below 2,000 meters, the degree of compensation might not be enough to ensure a protective effect, while above 3,500 meters, the adaptative compensatory mechanisms such as significant polycythemia and vascular stasis might increase the risk of thrombosis and, therefore, the risk of developing stroke [12,20,32] (Figure 2).

Figure 2 Proposed mechanisms and hypothesized physiopathology at low altitude (< 2,500 m), at high altitude (2,500-3,500 m) and very high altitude (>3,500 m). BP: Barometric pressure and O_2 is the oxygen availability in relationship to sea level FiO_2. : Angiogenesis occurs at different elevations above sea level but during the hypoxic beneficial window, polycythemia and red blood cell as well as platelet adhesiveness is not significant as above 3,500 m, thus, the protective effect reaches its maximum. Above 3,500 m although angiogenesis is present, the significantly high hematocrit and polycythemia increase the risk of blood stasis and thrombogenesis.

The information is still contradictory and opposed from one study to another. The few studies available have many limitations, and confounders' control was low in most of them. Nevertheless, very few studies that are better controlled and designed support some of our statements above. This report was designed to guide clinicians and researchers who are currently working with stroke and wanted to understand the role of

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elevation and hypobaric hypoxia for developing stroke while we suggest that further analysis and well-controlled studies are needed.

Limitations

Several limitations were found, including scarce information, conflicting results, and data lack when adjusting for confounders. In this sense, more research is needed to obtain a definitive answer; nevertheless, the information provided in this document can be used as an updated guide of the possible role of high-altitude exposure as a risk factor for developing a stroke.

Conclusions

It seems clear that short-term exposures to very high altitudes are a risk factor for developing a stroke, but when populations chronically reside between 2,000 to 3,500 meters, it seems like there is a protective effect, especially among native high-altitude dwellers. The available scientific literature suggests that above 3,500 to 4,000, the risk of developing stroke increases, especially if the exposure is acute among non-adapted populations.

When residing at moderate or high altitude (2,000-3,500 m), the suggestive protective effect might be due to increased perfusion within the tissues in response to hypoxia-triggered angiogenesis and vascular remodelling. Simultaneously, a hypercoagulable state might be responsible for the higher risk when exposed to very high or extreme altitudes.

Recommendations

Further investigations are needed to explore the role of socioeconomic variables and traditional risk factors among populations located at different elevations. Understanding the cultural and social differences between highlanders and lowlanders will permit more suitable and robust conclusions when elevation becomes a protective factor and when the parabola from lower risk to higher risk shifts above a certain altitude.

Abbreviations

ABP: Arterial Blood pressure
BMI: Body max index
BP: Barometric Pressure
COPD: Chronic obstructive pulmonary disease
CVD: Cerebro-vascular disease
CVT: Cerebro-venous thrombosis
CT: Computed tomography

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HS: haemorrhagic stroke

IHD: Ischemic heart diseases

INEC: Ecuadorian Institute of Statistics and Census

IS: Ischemic stroke

ICD-10: 10th revision of the International Statistical Classification of Diseases and Related Health Problems

I60: Nontraumatic subarachnoid haemorrhage

I61: Nontraumatic intracerebral haemorrhage

I62: Cerebral infarction

I64: Other and unspecified nontraumatic intracranial haemorrhage

MRI: Magnetic Resonance Imaging

SES: Socioeconomic status

Availability of supporting data

All the information used for this analysis can be found online throughout the several medical databases available

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

EOP, SPC, EV, GV and PC contributed with the review of the bibliography and preparation of the manuscript, EV and EOP were in charge of the entire revision of the document and finalizing the version end of our manuscript

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References

1 Hankey GJ. Stroke. *The Lancet* 2017;389:641–54. doi:10.1016/S0140-6736(16)30962-X

2 Wilson D, Charidimou A, Ambler G, *et al.* Recurrent stroke risk and cerebral microbleed burden in ischemic stroke and TIA: a meta-analysis. *Neurology* 2016;87:1501–10.

3 Seshadri S, Wolf PA. Lifetime risk of stroke and dementia: current concepts, and estimates from the Framingham Study. *Lancet Neurol* 2007;6:1106–14.

4 Johnson W, Onuma O, Owolabi M, *et al.* Stroke: a global response is needed. *Bull World Health Organ* 2016;94:634.

5 Fullerton HJ, Wu YW, Zhao S, *et al.* Risk of stroke in children: ethnic and gender disparities. *Neurology* 2003;61:189–94.

For peer review only

6 deVeber GA, Kirton A, Booth FA, *et al.* Epidemiology and outcomes of arterial ischemic stroke in children: the Canadian Pediatric Ischemic Stroke Registry. *Pediatr Neurol* 2017;69:58–70.

7 Bridgwood B, Lager KE, Mistri AK, *et al.* Interventions for improving modifiable risk factor control in the secondary prevention of stroke. *Cochrane Database Syst Rev* 2018.

8 Cheng Y-C, Ryan KA, Qadwai SA, *et al.* Cocaine use and risk of ischemic stroke in young adults. *Stroke* 2016;47:918–22.

9 Everson SA, Roberts RE, Goldberg DE, *et al.* Depressive symptoms and increased risk of stroke mortality over a 29-year period. *Arch Intern Med* 1998;158:1133–8.

10 Guzik A, Bushnell C. Stroke Epidemiology and Risk Factor Management. *Contin Lifelong Learn Neurol* 2017;23:15–39.

11 Szawarski P(1), Tam EWY(2), Richards P(3). Stroke at high altitude. *Hong Kong Med J* 2012;18:261.

12 Jha SK, Anand AC, Sharma V, *et al.* Stroke at high altitude: Indian experience. *High Alt Med Biol* 2002;3:21–7.

13 Niaz A, Nayyar S. Cerebrovascular stroke at high altitude. *J Coll Physicians Surg--Pak JCPSP* 2003;13:446–8.

14 Pilz S, Dobnig H, Fischer JE, *et al.* Low vitamin D levels predict stroke in patients referred to coronary angiography. *Stroke* 2008;39:2611–3.

15 Gürdal A, Keskin K, Orken DN, *et al.* Evaluation of Epicardial Fat Thickness in Young Patients With Embolic Stroke of Undetermined Source. *The neurologist* 2018;23:113–7.

16 Burtscher M. Effects of Living at Higher Altitudes on Mortality: A Narrative Review. *Aging Dis* 2014;5:274.

17 Chan T(1), Mak HKF(1), Wong WWY(2), *et al.* Acute ischaemic stroke during short-term travel to high altitude. *Hong Kong Med J* 2012;18:63–5.

18 Jaillard AS, Hommel M, Mazetti P. Prevalence of stroke at high altitude (3380 m) in Cuzco, a town of Peru. *Stroke* 1995;26:562–8.

19 Faeh D, Gutzwiller F, Bopp M. Lower mortality from coronary heart disease and stroke at higher altitudes in Switzerland. *Circulation* 2009;120:495–501. doi:10.1161/CIRCULATIONAHA.108.819250

20 Zangari M, Fink L, Tolomelli G, *et al.* Could hypoxia increase the prevalence of thrombotic complications in polycythemia vera? *BLOOD Coagul FIBRINOLYSIS* 2013;24:311–6. doi:10.1097/MBC.0b013e32835bfdb9

21 Gupta N, Ashraf MZ. Exposure to High Altitude: A Risk Factor for Venous Thromboembolism? *Semin Thromb Hemost* 2012;38:156–63. doi:10.1055/s-0032-1301413

22 Kotwal J(1), Apte CV(2), Kotwal A(3), *et al.* High altitude: A hypercoagulable state: Results of a prospective cohort study. *Thromb Res* 2007;120:391–7. doi:10.1016/j.thromres.2006.09.013

23 George J. The central asian expedition of capt. Roberovsky and lt. Kkozloff. 1896.

For peer review only

24 Lutsenko I. Prevalence of different stroke subtypes according toast criteria on high, medium and low altitude in Kyrgyzstan. *J Neurol Sci* 2017;381:147–147.

25 Al Tahan A, Buchur J, El Khwsky F, *et al.* Risk factors of stroke at high and low altitude areas in Saudi Arabia. *Arch Med Res* 1997;29:173–7.

26 Razdan S, Koul RL, Motta A, *et al.* Cerebrovascular disease in rural Kashmir, India. *Stroke* 1989;20:1691–3.

27 Annobil SH, Omojola MF, Adzaku FK, *et al.* Cerebrovascular accidents (strokes) in children with sickle cell disease residing at high and low altitudes of Saudi Aurabia. *Ann Trop Paediatr* 1990;10:191–8.

28 García J. Enfermedad cerebrovascular: factores de riesgo en los adultos nativos de la altura, HCRI-IPSS, 1989-1992. *Rev Peru EpidemiolOnline* 1994;7:67–72.

29 Mahajan SK, Kashyap R, Sood BR, *et al.* Stroke at moderate altitude. *JAPI* 2004;52:699–702.

30 Ezzati M, Horwitz ME, Thomas DS, *et al.* Altitude, life expectancy and mortality from ischaemic heart disease, stroke, COPD and cancers: national population-based analysis of US counties. *J Epidemiol Community Health* 2012;66:e17–e17.

31 Dhiman D, Mahajan SK, Sharma S, *et al.* The evolving pattern and outcome of stroke at moderate altitude. *J Neurosci Rural Pract* 2018;9:68.

32 Khattar NK, Sumardi F, Zemmar A, *et al.* Cerebral Venous Thrombosis at High Altitude: A Retrospective Cohort of Twenty-one Consecutive Patients. 2019.

33 West JB, Schoene RB, Milledge JS, *et al.* High altitude medicine and physiology. Hodder Arnold London 2007. http://www.jrnms.com/wp-content/uploads/2014/05/JRNMS-95-1-40-43.pdf (accessed 16 Aug 2016).

34 Imray C, Booth A, Wright A, *et al.* Acute altitude illnesses. *Bmj* 2011;343:d4943.

35 Brill A, Suidan GL, Wagner DD. Hypoxia, such as encountered at high altitude, promotes deep vein thrombosis in mice. *J Thromb Haemost JTH* 2013;11:1773–5. doi:10.1111/jth.12310

36 Khalil KF, Saeed W. Pulmonary Embolism in Soldiers Serving At High Altitude. *JCPSP-J Coll PHYSICIANS Surg Pak* 2010;20:468–71.

37 Kumar S(1 2). High altitude induced deep venous thrombosis: A study of 28 cases. *Indian J Surg* 2006;68:84–8.

38 Zavanone C, Panebianco M, Yger M, *et al.* Cerebral venous thrombosis at high altitude: A systematic review. *Rev Neurol (Paris)* 2017;173:189–93. doi:10.1016/j.neurol.2016.11.004

39 Azad P, Haddad GG. Molecular Basis of Hypoxia-Induced Excessive Erythrocytosis of High Altitude. In: *A74. RESPIRATORY HEALTH: ALTITUDE, ENVIRONMENTAL EXPOSURES, AND REHABILITATION*. American Thoracic Society 2018. A2366–A2366.

40 Steinback CD, Poulin MJ. Ventilatory responses to isocapnic and poikilocapnic hypoxia in humans. *Respir Physiol Neurobiol* 2007;155:104–13.

41 Ortiz-Prado E, Natah S, Srinivasan S, *et al.* A method for measuring brain partial pressure of oxygen in unanesthetized unrestrained subjects: the effect of acute and chronic hypoxia on brain tissue PO(2). *J Neurosci Methods* 2010;193:217–25. doi:10.1016/j.jneumeth.2010.08.019

For peer review only

42 Blitzer ML, Loh E, Roddy M-A, *et al.* Endothelium-derived nitric oxide regulates systemic and pulmonary vascular resistance during acute hypoxia in humans. *J Am Coll Cardiol* 1996;28:591–6.

43 Stenmark KR, Pugliese SC, Poth J, *et al.* The Effects of Chronic Hypoxia on Inflammation and Pulmonary Vascular Function. In: *Pulmonary Hypertension*. Springer 2016. 83–103.

44 Dunn JF, Wu Y, Zhao Z, *et al.* Training the Brain to Survive Stroke. *PLoS ONE* 2012;7:e45108. doi:10.1371/journal.pone.0045108

45 Bhalla IP, Naqui S, Ohri VC, *et al.* Lipid profile in polycythaemia of high altitude. *J Assoc Physicians India* 1988;36:374–5.

46 Brutsaert T. Why are high altitude natives so strong at high altitude? Nature vs. nurture: genetic factors vs. growth and development. In: *Hypoxia*. Springer 2016. 101–12.

47 Kollias J, Buskirk ER, Akers RF, *et al.* Work capacity of long-time residents and newcomers to altitude. *J Appl Physiol* 1968;24:792–9.

48 Moore LG. Measuring high-altitude adaptation. *J Appl Physiol* 2017;123:1371–85.

49 Vincent J, Hellot MF, Vargas E, *et al.* Pulmonary gas exchange, diffusing capacity in natives and newcomers at high altitude. *Respir Physiol* 1978;34:219–31.

50 Beall CM, Brittenham GM, Strohl KP, *et al.* Hemoglobin concentration of high-altitude Tibetans and Bolivian Aymara. *Am J Phys Anthropol* 1998;106:385–400.

51 Gassmann NN, van Elteren HA, Goos TG, *et al.* Pregnancy at high altitude in the Andes leads to increased total vessel density in healthy newborns. *J Appl Physiol* 2016;121:709–15.

52 Huicho L, Cabala-Peralta P, van-Elteren H-A, *et al.* Pregnancy at high altitude in the Andes leads to increased total vessel density in healthy newborns. 2016.

53 Chiodi H. Respiratory adaptations to chronic high altitude hypoxia. *J Appl Physiol* 1957;10:81–7.

54 León-Velarde F, Richalet J-P. Respiratory control in residents at high altitude: physiology and pathophysiology. *High Alt Med Biol* 2006;7:125–37.

55 Naeije R. Physiological adaptation of the cardiovascular system to high altitude. *Prog Cardiovasc Dis* 2010;52:456–66.

56 Ostadal B, Kolar F. Cardiac adaptation to chronic high-altitude hypoxia: beneficial and adverse effects. *Respir Physiol Neurobiol* 2007;158:224–36.

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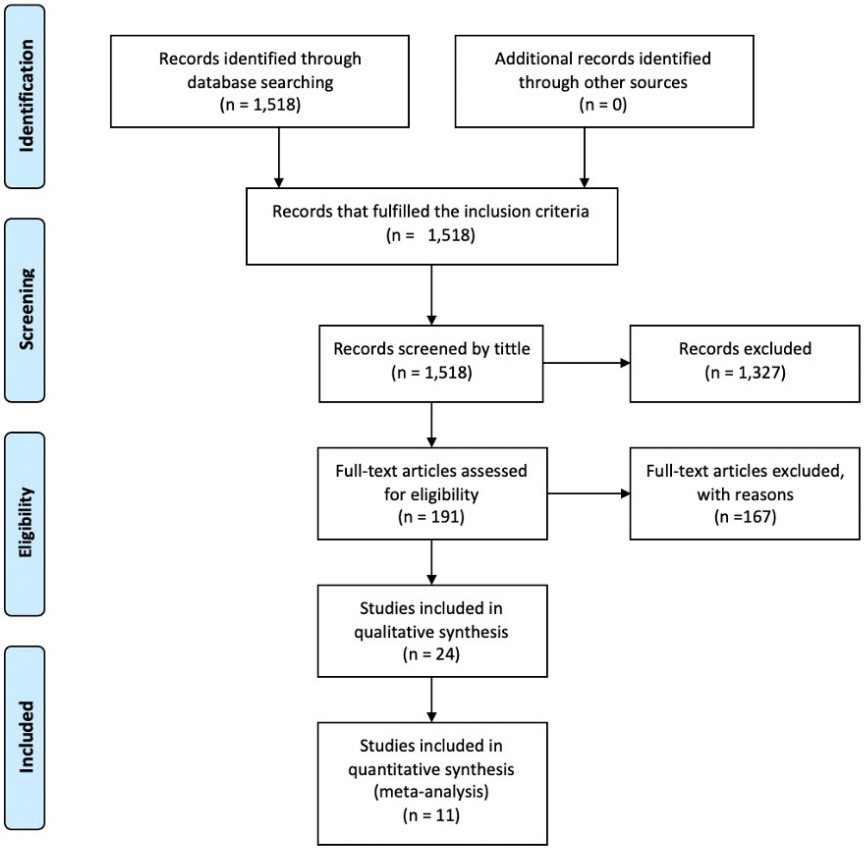


Figure 1

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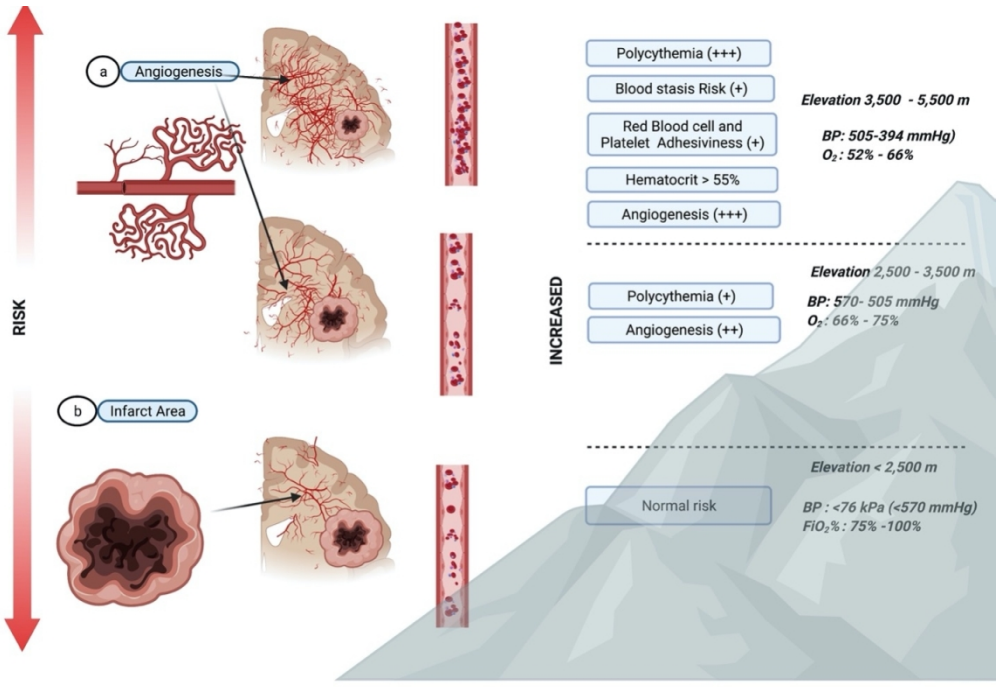


Figure 2

165x114mm (220 x 220 DPI)

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		Reporting Item	Page Number
Title			
	#1	Identify the report as a systematic review, meta-analysis, or both.	1
Abstract			
Structured summary	#2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	2
Introduction			
Rationale	#3	Describe the rationale for the review in the context of what is already known.	4

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Objectives	#4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
Methods			
Protocol and registration	#5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address) and, if available, provide registration information including the registration number.	4
Eligibility criteria	#6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rational	4
Information sources	#7	Describe all information sources in the search (e.g., databases with dates of coverage, contact with study authors to identify additional studies) and date last searched.	4
Search	#8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	#9	State the process for selecting studies (i.e., for screening, for determining eligibility, for inclusion in the systematic review, and, if applicable, for inclusion in the meta-analysis).	4
Data collection process	#10	Describe the method of data extraction from reports (e.g., piloted forms, independently by two reviewers) and any processes for obtaining and confirming data from investigators.	4
Data items	#11	List and define all variables for which data were sought (e.g., PICOS, funding sources), and any assumptions and simplifications made.	4
Risk of bias in individual studies	#12	Describe methods used for assessing risk of bias in individual studies (including specification of whether this was done at the study or outcome level, or both), and how this information is to be used in any data synthesis.	5
Summary measures	#13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Planned methods of analyis	#14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I2) for each meta-analysis.	5

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Risk of bias across studies	#15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	#16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
Results			
Study selection	#17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	#18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citation.	6
Risk of bias within studies	#19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see Item 12).	5
Results of individual studies	#20	For all outcomes considered (benefits and harms), present, for each study: (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot.	7-15
Synthesis of results	#21	Present the main results of the review. If meta-analyses are done, include for each, confidence intervals and measures of consistency.	7-15
Risk of bias across studies	#22	Present results of any assessment of risk of bias across studies (see Item 15).	7-15
Additional analysis	#23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7-15
Discussion			
Summary of Evidence	#24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., health care providers, users, and policy makers	16
Limitations	#25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	20
Conclusions	#26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	20

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Funding	#27	Describe sources of funding or other support (e.g., supply of data) for the systematic review; role of funders for the systematic review.	21

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Chronic High altitude exposure and the epidemiology of ischemic stroke, A systematic review

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Chronic High altitude exposure and the epidemiology of ischemic stroke, A systematic review

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Word count: 3,900

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Abstract

Introduction

About 5.7% of the world population resides above 1,500 m. It has been hypothesized that acute exposure to high-altitude locations can increase stroke risk, while chronic hypoxia can reduce stroke-related mortality.

Objective

This review aims to provide an overview of the available evidence on the association between long-term high altitude exposure and ischemic stroke.

Design

A Systematic literature review of the available information was performed from 1960 to 2021 to answer the link between hypobaric hypoxia due to high altitude exposure and stroke. The scientific repositories and databases were accessed using the University of Southampton library tool Delphis. In this review we included all scientific studies that were performed in humans living or visiting high altitude locations for at least 28 days as a cut-off point for chronic exposure.

Results

We reviewed a total of 1,890 abstracts retrieved during the first step of the literature review process. The authors acquired in full text as potentially relevant 204 studies. Only 17 documents met the inclusion criteria and were finally included. Ten studies clearly suggest that living at high altitudes may be associated with an increased risk of stroke, however, five studies suggest that altitude may act as a protective factor for the development of stroke, while two studies report ambiguous results.

Conclusions

This review suggest that the most robust studies tend to advocate that prolonged living at higher altitudes reduces the risk of developing stroke or dying from it. Increased irrigation due to angiogenesis and increased vascular perfusion might be the reason behind improved survival profiles among those living within this range. In contrast, residing above 3,500 m, seems to be associated with an apparent increased risk of developing stroke, probably linked to the presence of polycythaemia and other factors such as increased blood viscosity.

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Keywords: Stroke; High altitude; Hypoxia; Thrombosis; angiogenesis; Review

Strengths and limitations of this study

- This review analyzes the relationship between stroke and high altitude exposure using published data.
- Contains a quantitative analysis of the risk factors associated with ischemic stroke at high altitude.
- One of the main limitations is the lack of prospective cohort studies at the global level.
- The analysis excludes all case reports that exist in the scientific literature.

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Introduction

Stroke, also known as a cerebrovascular event, is a severe and often life-threatening medical condition that appears when the blood supply to part of the brain is interrupted [1]. It causes a rapid, acute, and potentially progressive brain function loss due to a disturbance in blood supply, oxygen, and nutrients that lasts at least 24 hours [2].

Cerebrovascular events or stroke is the second leading cause of death worldwide, affecting more than 16 million people each year[1]. Around 1 in 6 men and 1 in 5 women will have a stroke in their lifetime, and it is the third leading cause of disability worldwide [3,4]. Stroke affects people of all ages, though the causes of a stroke at a younger age are often different from those at older ages, especially in terms of risk factors and severity [5,6].

Clinical manifestations usually depend on the severity of the anoxia, the brain's location, the type and subtype of stroke, and the presence of other individual risk factors. The most common clinical manifestations are sudden unilateral weakness or numbness, diplopia, ataxia, and aphasia [1,7].

Stroke risk factors are often classified as traditional and non-traditional, and from this, they can be modifiable or non-modifiable [8]. In general terms, these risk factors have been classified as those which cause an ischemic stroke or haemorrhagic, being the most common and recognized as displayed in table 1.

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Table 1 Traditional modifiable and non-modifiable risk factors of stroke

Traditional risk factors			
		Non-modifiable	**Modifiable**
Ischemic Stroke	**Thrombotic**	Age Sex Ethnicity Socio-economic status Family History Arterial dissection	Waist circumference Alcohol misuse Obesity (BMI, WC, WHR) Diabetes Cigarette Smoking
	Embolic	Atrial fibrillation Intracardiac thrombus Heart valve disease Trauma and fractures Some types of Surgeries	Physical inactivity Apolipoprotein B to A1 Hyperlipidaemia***
	Systemic	Post-traumatic hypovolemia acute systemic hypoxia	
Haemorrhagic stroke	***ICH**	Vascular Malformations bleeding diatheses Trauma	Hypertension Cigarette smoking Obesity Waist -to-hip ratio
	****SAH**	Cocaine Abuse Amphetamines	Diet

*Intracerebral haemorrhage ** Subarachnoid haemorrhage *** High LDL serum levels

The risk of developing stroke increases with the presence of causal factors, which include arterial hypertension, atrial fibrillation, cigarette smoking, hyperlipidaemia, diabetes mellitus and acute infections such as COVID-19 [1,9]. Other modifiable factors are obesity, chronic kidney disease, excessive alcohol and cocaine consumption, sedentarism, psychological stress or depression [10–12]. Other factors such as Vitamin D or high-altitude may play a role but have been less studied than the traditional vascular risk factors [13–17].

Rationale for high altitude exposure and stroke

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Globally, at least 5.7% of the population lives above 1,500 meters, nevertheless, the association between high-altitude exposure and stroke has been poorly studied [14,15,18]. The available literature is based mainly on case reports due to acute exposure to extreme altitudes, rather than longer-term living among high-altitude dwellers [14,19–21]. The few reports available suggest that living in high-altitude regions (>2,500 m) increases the risk of developing thrombosis through hypoxia driven polycythaemia which leads to a hypercoagulation unbalance, that have been associated with increased risk for the development of atherothrombotic stroke [22–24].

Although most of the information comes from a case or case-series reports, no cohorts' studies have been published in this area. The only available information came from very few cross-sectional analysis that has found a significant association between living in high-altitude regions and having a greater risk of developing stroke, especially among younger populations [14,15,20,21].

Objective

To further explore this relationship, we have conducted a literature review of the available information in terms of the link between high-altitude chronic exposure and the risk of developing ischemic stroke.

Methods

Research Question

Does living at high-altitude increases the risk of developing ischemic stroke?

Study Design

A systematic review was conducted, including published scientific reports including original research, abstracts, editorial and commentary letters. We have used the preferred reporting guidelines for systematic reviews and meta-analyses (PRISMA). No protocol was registered for this review since PROSPERO does not currently accept registrations for scoping reviews, literature reviews or mapping reviews.

Data Source and search strategy

The literature review process was performed in English to cover the largest number of scientific databases and repositories containing academic literature from all over the world that are accessible for most researchers. The following libraries, repositories, and databases were accessed using the University of Southampton library tool Delphis: AMED, EMBASE, Cochrane Library, PubMed, MEDLINE, and Europe

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PMC. The Latin-American bibliographic database Scielo was also included in a separate search from its website repository.

The studies included in based on the following criteria:

Population: Adults from both sexes, all ages and ethnic background.

Study designs: All epidemiological analysis including observational and experimental full research, abstracts, editorials o commentary letters.

Exposure: We included every elevation above 2,000 m that was considered high altitude by the authors

Outcome of interest: Incidence and mortality rates

Inclusion criteria

This systematic literature review included all scientific papers that were written in English and reported epidemiological findings on the impact of stroke in different populations exposed to moderate to high altitudes. We set the cut-off point of short or long-term exposure in 28 days based on previously published data.

The main inclusion criteria were:

a) all manuscripts involving human subjects

b) all manuscripts that considered high altitude in the title or abstract and had the word stroke in the title.

c) all studies published between 1960 and 2021

The precise, full search strategy for all databases, registers, and websites, including any filters and limits used is described within the supplementary file 1.

Exclusion criteria

Exclusion criteria included all reports of thrombosis, deep vein thrombosis or venous thrombosis.

The list of the main exclusion criteria was:

a) Scientific papers concerning stroke that did not consider high altitude within the analysis.

b) This literature review excluded all in-vivo and in-vitro studies.

c) If the search term "stroke" referred to cardiac output or stroke volume, the results were excluded.

d) Case reports were excluded from the study.

Bias assessment

In order to reduce risk of any type of bias, the data extraction process was performed by EO and SC independently and in different times. in order to minimize errors while gathering information from any primary study, any disagreements was resolved after reaching consensus. In addition, while assessing the

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quality of primary studies, the CASP critical appraisal tool was used to reduce confirmation bias (supplementary file 2).

Data Synthesis

In this section we completely reviewed all the manuscripts that met the inclusion criteria. With the information obtained, a quantitative analysis of the results using the CASP tool was performed. The information coming from each of the manuscripts was synthesized in two tables.

Patient and Public Involvement

No patient involved

Results

Literature review

Several anecdotal case reports and case series of people suffering from stroke-related disorders at high-altitude have been published since the late 1800s [25]. Despite this apparent relationship, very few well-conducted epidemiological studies have been conducted worldwide, and we have only included those studies that fulfilled our inclusion and exclusion criteria [14,15,20,21,26–36].

We included 1,890 manuscripts during the first step of the literature review process, and only those that fulfil the inclusion criteria were revised. After this process, the authors evaluated 204 abstracts. In the end, we included 17manuscripts within this literature review (Figure 1).

Figure 1 Prisma chart diagram for reporting literature review search and metanalysis

Specific studies

Study 1: The first study published by Razdan et al. in 1988 looked into the prevalence of stroke in elevations above 1,500 m [26]. They found that the crude prevalence of stroke was higher than the populations residing

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below this elevation (Table 2). The study was carried out in India and includes 91 cases of stroke, the sample included men and women between 5 and 65 olds [26].

Study 2: In 1990, Annoml et al. published an article about the incidence of cerebrovascular accidents in children with sickle cell disease residing at high and low altitudes in Saudi Arabia. They found that 4% and 5% of the 400 children included in the study who were living at low and high altitudes developed stroke. The authors did not explore any causal relationship or controlled for confounders. The results presented in that study are not significant, considering the total number (n=8) of children with sickle cell diseases and stroke [27].

Study 3: In 1994, Garcia-J reported the prevalence of risk factors of developing stroke at 2,500 m in a community in Peru; nevertheless, the study focused on the association between risk factors and stroke and not in the relationship with elevation [28]. In this study, the author discusses cerebrovascular events in people living above 2,500 m but does not analyse the role of hypoxia as a triggering or protective factor. We came to this work because it was conducted at high altitudes and had stroke as an outcome, nevertheless, no comments about hypoxia were given in this study.

Study 4: In 1995, Jaillard et al. published one of the most cited stroke documents at high-altitude. A document from Perú described the results from a "door to door" survey in Cuzco, a city located at 3,380 meters above sea level. In this study, men and women over 15 years of age were included and classified in -97% mestizos, 1.3% white, 1.1% Quechua, 0.1% Aymara and 0.2% were from other racial groups. They reported a crude prevalence rate of 6.47 per 1,000 population; nevertheless, comparisons were performed using standard populations from other locations, while not giving enough information about the association of high-altitude and stroke per se. This study's evidence is limited since the comparison was performed with other reports at lower elevations; therefore, the conclusions should be taken cautiously [20].

Study 5: A Population-based analysis published by Jha et al. in 2002 reported a significantly higher incidence of stroke among patients admitted to a single tertiary hospital in India. They classified those patients that came from lower altitudes (<3,000 m) and higher altitudes (>3,000 m). They reported that the majority of stroke cases at high-altitude were presented in otherwise-healthy young men (<45 years old) exposed to high-altitude for months (nine months on average). The hospital admission rate at high-altitude was reported to be 12 times greater (12.8/1,000) than at lower altitudes (1.0/1,000), being this results statistically significant [14].

Study 6 In 2003, Niaz et al. conducted a case-control study to describe the prevalence of stroke above 4,571 m in India . They included 4,000 soldiers (20-40 y/o) from a military hospital to observe stroke occurrence. They reported 10 cases of stroke among young men patients from the high-altitude group

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versus only one case from the lower land cohort. Although no further information about size effect or standardization of the sample was provided, the researchers concluded that living at elevations above 4,500 m was associated with ten times more risk of developing stroke. The authors concluded that chronic (> 28 days) hypobaric hypoxia exposure it is associated with a higher risk of developing stroke, being massive ischemic stroke the most frequent presentation [15].

Study 7 In 2004, Mahajan et al. conducted a cross-sectional study in Himachal, Pradesh, India, among 100 patients between men and women some had risk factors such as hypertension or smoked from urban and rural areas hospitalized in a tertiary level hospital located at an altitude of 2200 meters. They reported that the incidence of stroke was lower when compared with previously published studies from India. Although information from the type of stroke and clinical features are available, the role of elevation was not discussed in depth. The authors compared the results with other studies, but no confounding analysis was made for the current report [29].

Study 8: An ecological study published by Faeh et al. in 2009 reported that stroke mortality is lower in higher locations in Switzerland (Faeh D et al., 2009). They reported that mortality from stroke decrease 12% per every 1000 m gained in elevation. The effect of altitude on the cardiovascular and cerebrovascular disease was assessed using sociodemographic information, places of birth and residence among men and women between 40 and 84 years of age living at altitudes of 259 to 1960 m. The protective effect of living at higher altitudes on coronary heart disease and stroke mortality was consistent and became stronger after adjustment for potential confounders [21].

Study 9: An ecological presented by Ezzati et al. in 2011, used from the National Elevation Dataset, National Centre for Health Statistics, and US Census. The authors analyzed the crude association of mean county altitude with life expectancy and mortality from ischaemic heart disease (IHD), stroke, chronic obstructive pulmonary disease (COPD), and cancers. Living at higher altitudes showed a beneficial association with ischemic heart disease mortality, but the results were not statistically significant for stroke [30].

Study 10: A prospectively collected Study in India published by Dhiman et al. 2018 reported the evolving Pattern and Outcome of Stroke at Moderate Altitude. This prospective, observational study was carried at 2,000 meters above sea level and included the clinical features and risk factors profile of 235 patients diagnosed with a stroke between nen and women with an average age of 62 years . The results only compared how the data compared with a report published 15 years ago, and the authors concluded that the occurrence of stroke decreased among hospitalized patients compared to the 2004 report [29,31].

Study 11: An Ecological study design in Nepal published by Khattar et al. in 2019 demonstrated that ischemic stroke is most likely to occur at elevations above 3,000 m and although this diagnosis is not

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technically considered stroke, the physiopathology of it might indicate some relationship with the development of ischemic cerebrovascular events. Despite some of these findings that targeted high-altitude as a possible risk factor for developing stroke, in 2009 [32] (Table 2). The study included 21 patients with stroke, 76% were men with a mean average age of 56 years, medical comorbidities included hypertension and diabetes were reported.

Study 12: An Hospital-based comparative study in Tibet and Beijing with 236 and 1021 in patients with acute ischemic stroke published by Lu et al in 2020 demonstrated that patients diagnosed with acute ischemic stroke located at 3650 m (Tibet) above sea level were younger than those who lived 40 m (Beijing) above sea level ($P < 0.001$); also, in Tibet, there was a higher proportion of cerebrovascular accidents in young adults, which may be associated with erythrocytosis [33].

Study 13: An ecological study published by Burtscher et al in 2021 carried out in Asutria demonstrated that mortality for all causes including stroke decrease as height increases, demonstrating a linear association; the mortality rate was lower around 15-20% in those who live at altitudes ranging from 1000 to 2000 m. It is important to note that the causes of death most affected by height are cardiovascular diseases and cancer, since at higher altitudes people have a lower mortality rate from these diseases[34].

Study 14: An ecological analysis of all stroke hospital admissions, mortality rates, and disability-adjusted life years in Ecuador published by Ortiz et al in 2021 demonstrated that high altitude populations had lower stroke mortality in men [OR: 0.91 (0.88–0.95)] and women [OR: 0.83 (0.79–0.86) and had a significant lower risk of getting admitted to the hospital when compared with the low altitude group in men [OR: 0.55 (CI 95% 0.54–0.56)] and women [OR: 0.65 (CI 95% 0.64–0.66)][35]

Study 15: Ecological Study carried out in China published by Liu et at demonstrated that Ischemic strokes at high altitudes were characterized by younger ages and larger infract volumes, also risk factors such as atherosclerosis, diabetes mellitus, coronary disease, and hyperlipidaemia were significantly lower than those in high-altitude areas [36].

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Table 2 Critical appraisal and summary of the literature.

Author and year	Tittle	Sample, setting and elevation	Aim	Design, type and length of Exposure	Main findings
Razdan et al. 1988	Cerebro-vascular Disease in Rural Kashmir, India	- The study was performed in Kuthar Valley, India -Elevation 1,530 m.	To find the prevalence and trends related to stroke in this region	-A cross sectional survey to find if population has suffered from stroke.	- 91 cases of stroke were detected - Crude prevalence rate 143/100,000 - Proportion in men (69,23%) and women (30,77%)
Annoml et.al 1990	Cerebrovascular accidents in children with sickle cell disease residing at high and low altitudes of Saudi Arabia	- The study was performed in Saudi Arabia -8 stroke cases -3,000 m	- To compare the clinical and radiological indications of stroke in sickle cell patients constantly residing at low and high-altitudes.	-Analyses of 8 cases of stroke among children with sickle cell disease -extracted from 400 cases detected in Saudi Arabia.	-The incidence of stroke in children with sickle cell disease similar regardless elevation.
Garcia, J 1994	Cerebrovascular Disease: Risk Factors analysis of Natives living at high-altitude.	-The study was performed in Peru -50 patients with a confirmed diagnosis of stroke and 52 healthy patients -2,500 m	To understand the role of different risk factors in patients located at high altitudes	Retrospective case-control study - Population selection based on criteria (cases), random population (controls)	- Atherosclerosis was the most important risk factor for CVD in the highland natives.
Jaillard, 1995	Prevalence of Stroke at High-altitude (3380 m) in Cuzco, a Town of Peru A Population-Based Study	- The study was performed in Peru -3,246 subjects interviewed -3,800 m	to study the prevalence of stroke and its association with stroke risk factors in a high-altitude location.	A door to door population-based analysis using a survey as a tool for collecting data.	-Age, polycythemia, high consumption of alcohol, and area of residence were associated with higher stroke prevalence.
Jha et. al 2002	Stroke at High-altitude: Indian Experience	- The study was performed in India -4,270 m	-To determine the relationship between stroke and high-altitude.	-Information was prospectively collected from the Command Hospital in India	-Long-term stay at high-altitude was associated with higher risk of developing stroke.
Niaz et al. 2003	Stroke at high altitude: Indian experience	-The study was performed in Chandimandir, India -Elevation: 4,200 m.	-To identify the presence of stroke using different clinical, imaging and laboratory assessments.	-A cohort analysis of stroke prevalence and mortality from 1998-2000.	-Stroke hospital admission rates was higher among high altitude residents (13.7/1000) versus low altitude (1.05/1000)
Mahajan et. al 2004	Stroke at moderate altitude	-this study was performed in Himachal, Pradesh in the Sub-Himalayan ranges. -elevation: 2,200 m	To understand the clinical profile, presence of various risk factors for stroke at moderate altitude.	-A cross sectional analysis of stroke admission in a tertiary hospital from India	-Patients aged up to 45 years were defined as stroke in young. -Complete clinical, radiological and neurological examination was performed.
Faeh et.al 2009	Lower Mortality from Coronary Heart Disease and Stroke at Higher Altitudes in Switzerland.	-The study was performed in Switzerland. - 1 641 144 Swiss and foreign nationals born in Switzerland -259 to 1960 m.	Examine mortality from coronary heart disease (CHD) and stroke in its association with high-altitude living.	-A longitudinal analysis of mortality data from 1990 to 2000.	-Living at higher altitude was associated with less CHD and less stroke mortality.
Ezzati et.al 2011	Altitude, life expectancy and mortality from ischaemic heart disease, stroke, COPD and cancers: national population-based analysis of US counties	The Study was performed in USA -Mortality and population data for 2001 through 2005. -12.1 million death records -500-1000 m -1000- 1500 m -1500 > m	Inspect the relationship of life expectancy and mortality from particular diseases in relationship to altitude.	-Ecological study of the association of mean county altitude with life expectancy and mortality from IHD, Stroke, COPD and cancer	- Counties above 1500 m had longer life expectances than those within 100m of sea level by 1.2 - 3.6 years for men and 0.5 -2.5 years for women.

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Dhiman et. al 2018	The Evolving Pattern and Outcome of Stroke at Moderate Altitude	- The study was performed in India -Men and women with an average age of 62 years Sub-Himalayan region of India. -235 patients with stroke were consecutively admitted into a tertiary hospital. -2,000 m	The aim was to compare the clinical profile, risk factors, and outcome in hospitalized patients of stroke in a Tertiary Care Hospital situated at moderate altitude	-A prospectively collected Study in India. -A comparative analysis was performed with a previous study performed 15 years prior of the current study.	-Ischemic stroke was noted in 74%, and 26% had haemorrhagic stroke (HS). -Male accounted 58% of the cases and female for 42%. -Overall HS had poorer outcome. -There occurrence of stroke has decreased among hospitalized patients at moderate altitude.
Khattar et.al 2019	Cerebral Venous Thrombosis at High-altitude: A Retrospective Cohort of Twenty-one Consecutive Patients	-The study cohort comprised 21 patients in Nepal. CVT was the diagnosis of interest. -elevation 3,048 m	Investigate the characteristics and treatment outcomes of patients who suffered CVT at high-altitude in eastern Nepal.	- Retrospectively reviewed all patients presenting with clinical and radiographic evidence of cerebral venous sinus thrombosis treated.	21 patients of which 76% were men with an average age of 56. -men were found to have a higher risk for CVT at high-altitude -All patients presented with evidence of haemorrhagic conversion on the initial brain CT.
Lu et.al 2020	Characteristics of acute ischemic stroke in hospitalized patients in Tibet: a retrospective comparative study	- The study was performed in Tibet and Beijing - 236 and 1021 inpatients with acute ischemic stroke from People's Hospital of Tibet Autonomous Region and Peking University First Hospital -40 m and 3650 m above sea level	This study aimed to analyse the clinical characteristics of AIS patients at high-altitude regions through a hospital-based comparative study between Tibet and Beijing.	-Hospital-based comparative study	- In Tibet, AIS patients were relatively younger, and anterior circulation infarctions were more common - Young adult stroke, erythrocytosis, and hyperhomocysteinemia were more frequent in PHOTAR patients
Burtscher et. al 2021	Does living at moderate altitudes in Austria affect mortality rates of various causes? An ecological study	- The study was performed in Austria - 467 834 deaths -Elevations: <251 m, 251–500 m, 501–750 m, , 751–1000, 1001-2000 m	This study tested the hypothesis that living at moderate altitudes (up to 2000 m) is associated with reduced mortality from all causes.	An ecological study	- ASMRs for residents living in higher (>1000 m) versus lower (<251 m) altitude regions (with agriculture employment below 3%) were 485.8 versus 597.0 (rate ratio and 95% CI 0.81 (0.72 to 0.92); p<0.001) for men and 284.6 versus 365.5 (0.78; 0.66 to 0.91); p=0.002) for women.
Ortiz et.al 2021	Stroke-Related Mortality at Different Altitudes: A 17-Year Nationwide Population-Based Analysis From Ecuador	- The study was performed in Ecuador - A total of 38,201 deaths and 75,893 hospital admissions due to stroke -Elevations low altitude (<1,500 m), moderate altitude (1,500–2,500 m), high altitude (2,500–3,500 m) and very high altitude (3,500–5,500 m)	Elucidate the association between stroke and altitude using four different elevation ranges	-An ecological analysis of all stroke hospital admissions, mortality rates, and disability-adjusted life years in Ecuador was performed from 2001 to 2017	- A total of 38,201 deaths and 75,893 stroke-related hospital admissions were reported -High altitude populations (HAP) had lower stroke mortality in men [OR: 0.91 (0.88–0.95)] and women [OR: 0.83 (0.79–0.86)].
Liu et.al 2021	Acute Ischemic Stroke at High Altitudes in China: Early Onset and Severe Manifestations	-The study was performed in China -892 cases of first-ever acute ischemic strokes at altitudes of 20, 2550, and 4200 m in China (697 cases from Penglai, 122 cases from Huzhu, and 73 cases from Yushu).	Describe and investigate the characteristics of acute ischemic stroke at different altitudes in China	Ecological Study	-Ischemic strokes at high altitudes were characterized by younger ages (69.14 ± 11.10 vs. 64.44 ± 11.50 vs. 64.45 ± 14.03, p < 0.001) and larger infract volumes (8436.37 ± 29,615.07 mm3 vs. 17,213.16 ± 47,044.74 mm3 vs. 42,459 ± 84,529.83 mm3, p < 0.001

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Discussion

The results from this review suggest that stroke seems to be more likely to occur in very high-altitude locations (>3,500 m) when the exposure is longer than 28 days, especially among younger people (< 45 y/o). On the other hand, when people live above 1,500 m and below 3,500 seems like there is a protective effect for stroke, probably triggered by better adaptation to hypoxia, efficient enough to reduce the likelihood of dying when compared to lower altitudes; nevertheless, no information is available about the exact point in which this protective effect become a risk factor.

It has been challenging to define how high altitude exposure can be defined and where the threshold is located in terms of mild or severe hypoxia [37]. For instance, Imray in 2017uses a classification of high-altitude exposure according to the recommendations from the International Society of Mountain Medicine, a categorization that seems to be the most pragmatic [38]. The author defined low altitude everything located below 1,500 m, moderate or intermediate altitude between 1,500 to 2,500 m, high-altitude from 2,500 to 3,500 m, the very high-altitude from 3,500 m to 5,800, more than 5,800 extreme high-altitudes and above the 8,000 m is considered the death zone [38].

As humans acclimatize to high-altitude, adverse and often mild secondary effects can occur in response to hypoxia. Some of these adverse effects are linked to the increased blood viscosity due to polycythemia, augmented pulmonary arterial pressure, and sometimes, they are linked to a proposed hypercoagulation unbalance [22,24].

These consequences might be increasing the risk of forming an atherothrombotic plaque resulting in a stroke or myocardial infarction (MI) or venous-thrombotic events, resulting in deep venous thrombosis (DVT) or pulmonary embolism (PE) [39–41]. Although information about the time of exposure is scarce, the longer the exposure, the higher the risk [21,42].

Acute exposure to hypobaric hypoxia produces several compensatory physiological effects that can last hours, days, months, or years. The essential mechanisms are: increasing the heart and respiratory rates, a secondary polycythemia, haemoconcentration derived from reduced plasma volume caused by respiratory evaporative water loss and polyuria, and increased ventilatory response [37,43,44]. When acute exposure last longer than 28 days, more efficient and prolonged mechanisms take place, including sustained polycythemia, endothelium changes, reduced vascular resistance, nitric oxide-mediated hypotension, and angiogenesis [45–48]. Acute exposure to high-altitude hypoxia triggers a series of events that produce a hypercoagulable state [24]. This hypercoagulability state is boosted by dehydration, haemoconcentration, and polycythemia. When combined with dehydration (due to tachyphemia and extenuate physical activity) and limited mobilization (sleeping in tents and secluded spaces), these factors produce the perfect scenario for increasing vascular stasis and thrombosis [22,37,49].

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When humans are exposed continuously to hypoxia, they develop adaptative mechanisms that are far more efficient than those observed in newcomers [50–53]. These last-longing mechanisms include anatomical (wider chests, shorter and lighter bodies, etc.), embryological (smaller fetus and placentas), circulatory (improved maximum flow output and higher pulmonary arterial pressure), and respiratory adaptations (improved hypoxic ventilatory response and oxygen diffusion capacities) [52,54–56]. Chronic exposure to hypobaric hypoxia leads to the development of more subtle compensatory mechanisms. These factors include long-term erythrocytosis, angiogenesis, capillary remodelling, and an improved ventilatory response [57–60] (Figure 2).

Once the general context of acute or chronic hypobaric hypoxia has been described, the main intrigue is which elevation is enough to generate compensatory mechanisms capable of reducing the risk of developing stroke and when these mechanisms become detrimental. After reviewing the current literature, the information available suggests that a window around 2,000 to 3,500 meters of elevation might be enough to generate some protective mechanisms (i.e., angiogenesis or vascular remodelling) against stroke [21,45,48].

In elevations below 2,000 meters, the degree of compensation might not be enough to ensure a protective effect, while above 3,500 meters, the adaptative compensatory mechanisms such as significant polycythemia and vascular stasis might increase the risk of thrombosis and, therefore, the risk of developing stroke [14,22,32] (Figure 2).

Figure 2 Proposed mechanisms and hypothesized physiopathology at low altitude (< 2,500 m), at high altitude (2,500-3,500 m) and very high altitude (>3,500 m). BP: Barometric pressure and O_2 is the oxygen availability in relationship to sea level FiO_2. : Angiogenesis occurs at different elevations above sea level but during the hypoxic beneficial window, polycythemia and red blood cell as well as platelet adhesiveness is not significant as above 3,500 m, thus, the protective effect reaches its maximum. Above 3,500 m although angiogenesis is present, the significantly high hematocrit and polycythemia increase the risk of blood stasis and thrombogenesis.

The information is still contradictory and opposed from one study to another. The few studies available have many limitations, and confounders' control was low in most of them. Nevertheless, very few studies that are better controlled and designed support some of our statements above. This report was designed to guide clinicians and researchers who are currently working with stroke and wanted to understand the role of

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elevation and hypobaric hypoxia for developing stroke while we suggest that further analysis and well-controlled studies are needed.

Limitations

Several limitations were found, including scarce information, conflicting results, and data lack when adjusting for confounders. In this sense, more research is needed to obtain a definitive answer; nevertheless, the information provided in this document can be used as an updated guide of the possible role of high-altitude exposure as a risk factor for developing a stroke.

Conclusions

This review suggest that the most robust studies tend to advocate that prolonged living at higher altitudes reduces the risk of developing stroke or dying from it. Increased irrigation due to angiogenesis and increased vascular perfusion might be the reason behind improved survival profiles among those living within this range. In contrast, residing high altitude locations, especially above 3,500 m, is associated with an apparent increased risk, probably linked to the presence of polycythaemia and other factors such as increased blood viscosity and the presence of a proposed hypercoagulable state might increase the risk of developing stroke among those exposed to very high altitudes. It seems clear that short-term exposures to very high altitudes are a risk factor for developing a stroke. The available scientific literature suggests that above 3,500 to 4,000, the risk of developing stroke increases, especially if the exposure is acute among non-adapted populations.

It is important to note that one of the main limitations presented by some of the studies analysed was the lack of analysis of risk factors related to stroke, in addition, the level of education, socioeconomic level, or living conditions of the participants were not analysed. We also highlight that certain risk factors such as diabetes, arteriosclerosis, coronary heart disease, or hyperlipidaemia have a lower prevalence in people living in high altitude areas

Recommendations

Further investigations are needed to explore the role of socioeconomic variables and traditional risk factors among populations located at different elevations. Understanding the cultural and social differences between highlanders and lowlanders will permit more suitable and robust conclusions when elevation becomes a protective factor and when the parabola from lower risk to higher risk shifts above a certain altitude.

Abbreviations

ABP: Arterial Blood pressure

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BMI: Body max index

BP: Barometric Pressure

COPD: Chronic obstructive pulmonary disease

CVD: Cerebro-vascular disease

CVT: Cerebro-venous thrombosis

CT: Computed tomography

HS: haemorrhagic stroke

IHD: Ischemic heart diseases

INEC: Ecuadorian Institute of Statistics and Census

IS: Ischemic stroke

ICD-10: 10th revision of the International Statistical Classification of Diseases and Related Health Problems

I60: Nontraumatic subarachnoid haemorrhage

I61: Nontraumatic intracerebral haemorrhage

I62: Cerebral infarction

I64: Other and unspecified nontraumatic intracranial haemorrhage

MRI: Magnetic Resonance Imaging

SES: Socioeconomic status

Availability of supporting data

All the information used for this analysis can be found online throughout the several medical databases available

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

EOP, SPC, EV, GV and PC contributed with the review of the bibliography and preparation of the manuscript, EV and EOP oversaw the entire revision of the document and finalizing the version end of our manuscript

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Ethics Statement

This secondary data analysis of publicly available anonymized data received ethical approval from the University of Southampton with the Faculty of Medicine Ethics Committee ERGO 51422.R3 number.

References

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1 Hankey GJ. Stroke. *The Lancet* 2017;**389**:641–54. doi:10.1016/S0140-6736(16)30962-X

2 Wilson D, Charidimou A, Ambler G, *et al.* Recurrent stroke risk and cerebral microbleed burden in ischemic stroke and TIA: a meta-analysis. *Neurology* 2016;**87**:1501–10.

3 Seshadri S, Wolf PA. Lifetime risk of stroke and dementia: current concepts, and estimates from the Framingham Study. *Lancet Neurol* 2007;**6**:1106–14.

4 Johnson W, Onuma O, Owolabi M, *et al.* Stroke: a global response is needed. *Bull World Health Organ* 2016;**94**:634.

5 Fullerton HJ, Wu YW, Zhao S, *et al.* Risk of stroke in children: ethnic and gender disparities. *Neurology* 2003;**61**:189–94.

6 deVeber GA, Kirton A, Booth FA, *et al.* Epidemiology and outcomes of arterial ischemic stroke in children: the Canadian Pediatric Ischemic Stroke Registry. *Pediatr Neurol* 2017;**69**:58–70.

7 Coleman ER, Moudgal R, Lang K, *et al.* Early rehabilitation after stroke: a narrative review. *Curr Atheroscler Rep* 2017;**19**:1–12.

8 Bridgwood B, Lager KE, Mistri AK, *et al.* Interventions for improving modifiable risk factor control in the secondary prevention of stroke. *Cochrane Database Syst Rev* 2018.

9 Nannoni S, de Groot R, Bell S, *et al.* Stroke in COVID-19: a systematic review and meta-analysis. *Int J Stroke* 2021;**16**:137–49.

10 Cheng Y-C, Ryan KA, Qadwai SA, *et al.* Cocaine use and risk of ischemic stroke in young adults. *Stroke* 2016;**47**:918–22.

11 Everson SA, Roberts RE, Goldberg DE, *et al.* Depressive symptoms and increased risk of stroke mortality over a 29-year period. *Arch Intern Med* 1998;**158**:1133–8.

12 Guzik A, Bushnell C. Stroke Epidemiology and Risk Factor Management. *Contin Lifelong Learn Neurol* 2017;**23**:15–39.

13 Szawarski P(1), Tam EWY(2), Richards P(3). Stroke at high altitude. *Hong Kong Med J* 2012;**18**:261.

14 Jha SK, Anand AC, Sharma V, *et al.* Stroke at high altitude: Indian experience. *High Alt Med Biol* 2002;**3**:21–7.

15 Niaz A, Nayyar S. Cerebrovascular stroke at high altitude. *J Coll Physicians Surg--Pak JCPSP* 2003;**13**:446–8.

16 Pilz S, Dobnig H, Fischer JE, *et al.* Low vitamin D levels predict stroke in patients referred to coronary angiography. *Stroke* 2008;**39**:2611–3.

17 Gürdal A, Keskin K, Orken DN, *et al.* Evaluation of Epicardial Fat Thickness in Young Patients With Embolic Stroke of Undetermined Source. *The neurologist* 2018;**23**:113–7.

18 Burtscher M. Effects of Living at Higher Altitudes on Mortality: A Narrative Review. *Aging Dis* 2014;**5**:274.

For peer review only

19 Chan T(1), Mak HKF(1), Wong WWY(2), *et al.* Acute ischaemic stroke during short-term travel to high altitude. *Hong Kong Med J* 2012;**18**:63–5.

20 Jaillard AS, Hommel M, Mazetti P. Prevalence of stroke at high altitude (3380 m) in Cuzco, a town of Peru. *Stroke* 1995;**26**:562–8.

21 Faeh D, Gutzwiller F, Bopp M. Lower mortality from coronary heart disease and stroke at higher altitudes in Switzerland. *Circulation* 2009;**120**:495–501. doi:10.1161/CIRCULATIONAHA.108.819250

22 Zangari M, Fink L, Tolomelli G, *et al.* Could hypoxia increase the prevalence of thrombotic complications in polycythemia vera? *BLOOD Coagul FIBRINOLYSIS* 2013;**24**:311–6. doi:10.1097/MBC.0b013e32835bfdb9

23 Gupta N, Ashraf MZ. Exposure to High Altitude: A Risk Factor for Venous Thromboembolism? *Semin Thromb Hemost* 2012;**38**:156–63. doi:10.1055/s-0032-1301413

24 Kotwal J, Apte CV, Kotwal A, *et al.* High altitude: A hypercoagulable state: Results of a prospective cohort study. *Thromb Res* 2007;**120**:391–7. doi:10.1016/j.thromres.2006.09.013

25 George J. The central asian expedition of capt. Roberovsky and lt. Kkozloff. 1896.

26 Razdan S, Koul RL, Motta A, *et al.* Cerebrovascular disease in rural Kashmir, India. *Stroke* 1989;**20**:1691–3.

27 Annobil SH, Omojola MF, Adzaku FK, *et al.* Cerebrovascular accidents (strokes) in children with sickle cell disease residing at high and low altitudes of Saudi Aurabia. *Ann Trop Paediatr* 1990;**10**:191–8.

28 García J. Enfermedad cerebrovascular: factores de riesgo en los adultos nativos de la altura, HCRI-IPSS, 1989-1992. *Rev Peru EpidemiolOnline* 1994;**7**:67–72.

29 Mahajan SK, Kashyap R, Sood BR, *et al.* Stroke at moderate altitude. *JAPI* 2004;**52**:699–702.

30 Ezzati M, Horwitz ME, Thomas DS, *et al.* Altitude, life expectancy and mortality from ischaemic heart disease, stroke, COPD and cancers: national population-based analysis of US counties. *J Epidemiol Community Health* 2012;**66**:e17–e17.

31 Dhiman D, Mahajan SK, Sharma S, *et al.* The evolving pattern and outcome of stroke at moderate altitude. *J Neurosci Rural Pract* 2018;**9**:68.

32 Khattar NK, Sumardi F, Zemmar A, *et al.* Cerebral Venous Thrombosis at High Altitude: A Retrospective Cohort of Twenty-one Consecutive Patients. 2019.

33 Lu Y, Zhuoga C, Jin H, *et al.* Characteristics of acute ischemic stroke in hospitalized patients in Tibet: a retrospective comparative study. *BMC Neurol* 2020;**20**:380. doi:10.1186/s12883-020-01957-0

34 Burtscher J, Millet GP, Burtscher M. Does living at moderate altitudes in Austria affect mortality rates of various causes? An ecological study. *BMJ Open* 2021;**11**:e048520. doi:10.1136/bmjopen-2020-048520

35 Ortiz-Prado E, Espinosa PS, Borrero A, *et al.* Stroke-Related Mortality at Different Altitudes: A 17-Year Nationwide Population-Based Analysis From Ecuador. *Front Physiol* 2021;**12**:1680. doi:10.3389/fphys.2021.733928

36 Liu M, Yan M, Guo Y, *et al.* Acute Ischemic Stroke at High Altitudes in China: Early Onset and Severe Manifestations. *Cells* 2021;**10**:809. doi:10.3390/cells10040809

For peer review only

37 West JB, Schoene RB, Milledge JS, *et al.* High altitude medicine and physiology. Hodder Arnold London 2007. http://www.jrnms.com/wp-content/uploads/2014/05/JRNMS-95-1-40-43.pdf (accessed 16 Aug 2016).

38 Imray C, Booth A, Wright A, *et al.* Acute altitude illnesses. *Bmj* 2011;**343**:d4943.

39 Brill A, Suidan GL, Wagner DD. Hypoxia, such as encountered at high altitude, promotes deep vein thrombosis in mice. *J Thromb Haemost JTH* 2013;**11**:1773–5. doi:10.1111/jth.12310

40 Khalil KF, Saeed W. Pulmonary Embolism in Soldiers Serving At High Altitude. *JCPSP-J Coll PHYSICIANS Surg Pak* 2010;**20**:468–71.

41 Kumar S(1 2). High altitude induced deep venous thrombosis: A study of 28 cases. *Indian J Surg* 2006;**68**:84–8.

42 Zavanone C, Panebianco M, Yger M, *et al.* Cerebral venous thrombosis at high altitude: A systematic review. *Rev Neurol (Paris)* 2017;**173**:189–93. doi:10.1016/j.neurol.2016.11.004

43 Azad P, Haddad GG. Molecular Basis of Hypoxia-Induced Excessive Erythrocytosis of High Altitude. In: *A74. RESPIRATORY HEALTH: ALTITUDE, ENVIRONMENTAL EXPOSURES, AND REHABILITATION*. American Thoracic Society 2018. A2366–A2366.

44 Steinback CD, Poulin MJ. Ventilatory responses to isocapnic and poikilocapnic hypoxia in humans. *Respir Physiol Neurobiol* 2007;**155**:104–13.

45 Ortiz-Prado E, Natah S, Srinivasan S, *et al.* A method for measuring brain partial pressure of oxygen in unanesthetized unrestrained subjects: the effect of acute and chronic hypoxia on brain tissue PO(2). *J Neurosci Methods* 2010;**193**:217–25. doi:10.1016/j.jneumeth.2010.08.019

46 Blitzer ML, Loh E, Roddy M-A, *et al.* Endothelium-derived nitric oxide regulates systemic and pulmonary vascular resistance during acute hypoxia in humans. *J Am Coll Cardiol* 1996;**28**:591–6.

47 Stenmark KR, Pugliese SC, Poth J, *et al.* The Effects of Chronic Hypoxia on Inflammation and Pulmonary Vascular Function. In: *Pulmonary Hypertension*. Springer 2016. 83–103.

48 Dunn JF, Wu Y, Zhao Z, *et al.* Training the Brain to Survive Stroke. *PLoS ONE* 2012;**7**:e45108. doi:10.1371/journal.pone.0045108

49 Bhalla IP, Naqui S, Ohri VC, *et al.* Lipid profile in polycythaemia of high altitude. *J Assoc Physicians India* 1988;**36**:374–5.

50 Brutsaert T. Why are high altitude natives so strong at high altitude? Nature vs. nurture: genetic factors vs. growth and development. In: *Hypoxia*. Springer 2016. 101–12.

51 Kollias J, Buskirk ER, Akers RF, *et al.* Work capacity of long-time residents and newcomers to altitude. *J Appl Physiol* 1968;**24**:792–9.

52 Moore LG. Measuring high-altitude adaptation. *J Appl Physiol* 2017;**123**:1371–85.

53 Vincent J, Hellot MF, Vargas E, *et al.* Pulmonary gas exchange, diffusing capacity in natives and newcomers at high altitude. *Respir Physiol* 1978;**34**:219–31.

54 Beall CM, Brittenham GM, Strohl KP, *et al.* Hemoglobin concentration of high-altitude Tibetans and Bolivian Aymara. *Am J Phys Anthropol* 1998;**106**:385–400.

For peer review only

55 Gassmann NN, van Elteren HA, Goos TG, *et al.* Pregnancy at high altitude in the Andes leads to increased total vessel density in healthy newborns. *J Appl Physiol* 2016;**121**:709–15.

56 Huicho L, Cabala-Peralta P, van-Elteren H-A, *et al.* Pregnancy at high altitude in the Andes leads to increased total vessel density in healthy newborns. 2016.

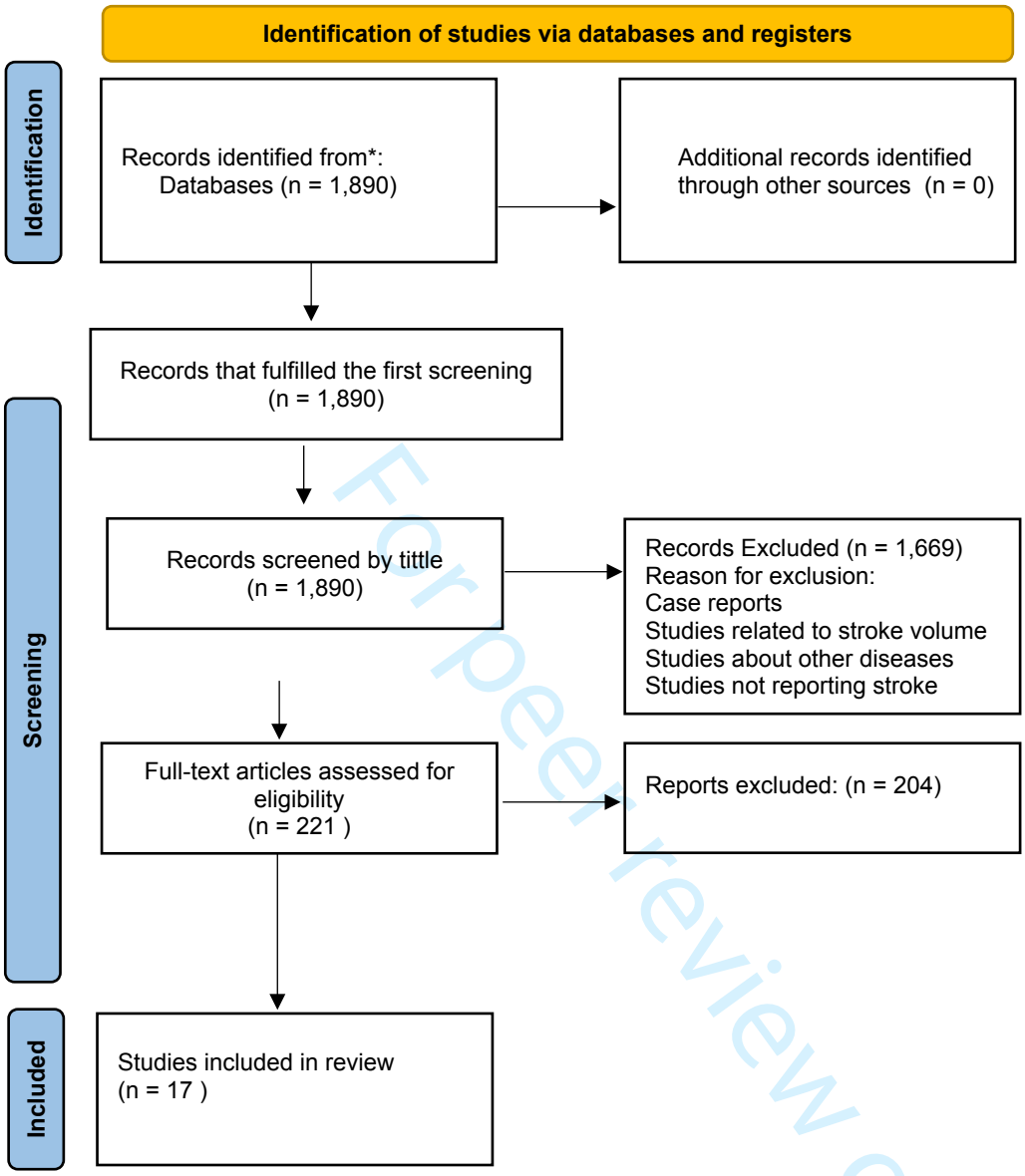
57 Chiodi H. Respiratory adaptations to chronic high altitude hypoxia. *J Appl Physiol* 1957;**10**:81–7.

58 León-Velarde F, Richalet J-P. Respiratory control in residents at high altitude: physiology and pathophysiology. *High Alt Med Biol* 2006;**7**:125–37.

59 Naeije R. Physiological adaptation of the cardiovascular system to high altitude. *Prog Cardiovasc Dis* 2010;**52**:456–66.

60 Ostadal B, Kolar F. Cardiac adaptation to chronic high-altitude hypoxia: beneficial and adverse effects. *Respir Physiol Neurobiol* 2007;**158**:224–36.

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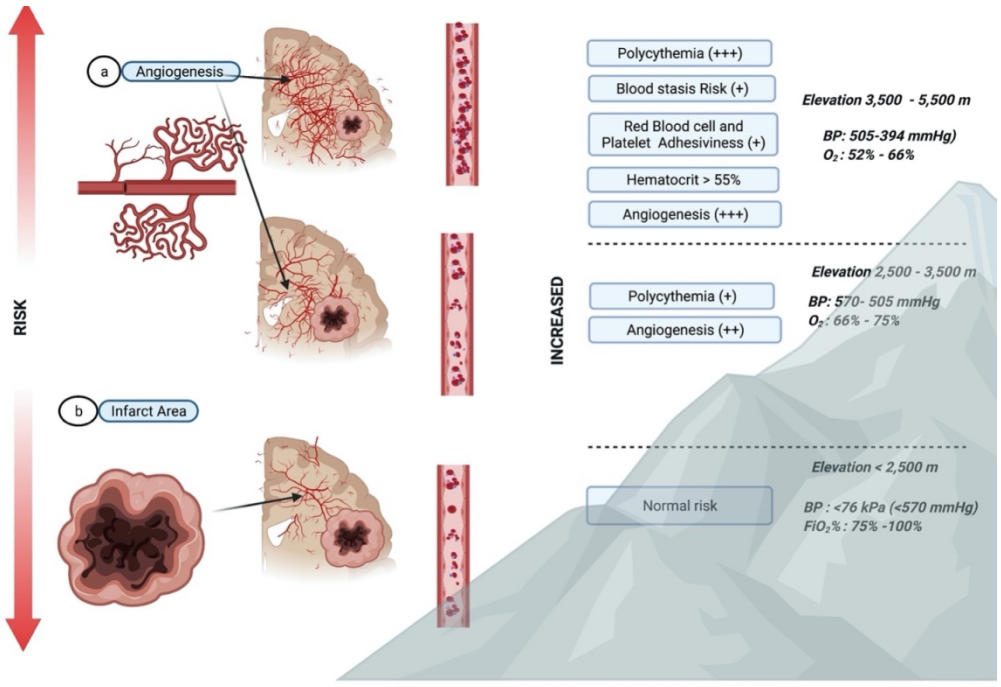


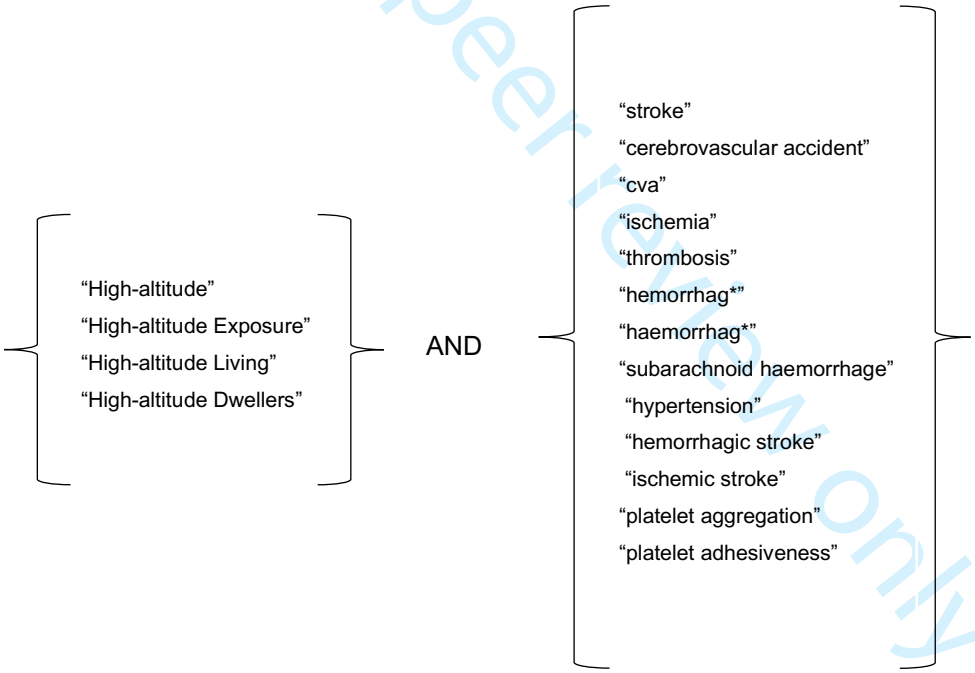
Figure 2

165x114mm (220 x 220 DPI)

Supplementary File 1

The literature review process was performed in English in order to cover the largest number of scientific databases and repositories containing academic literature from all over the world that are accessible for most of the researchers.

The following libraries, repositories and databases were accessed using the University of Southampton library tool Delphis: AMED, EMBASE, Cochrane Library, PubMed, MEDLINE and Europe PMC. The Latin-American bibliographic database Scielo was also included in a separate search from its website repository. The following terms were used during the search strategy while defined terms within the parenthesis were combined with the Boolean operator "OR".



The exclusion criteria for the literature review excluded all *in vivo* and *in vitro* studies and all the data analysed that did not concern human studies. If the search term "stroke" referred to cardiac output or stroke volume, the results were also excluded. The length of the data extraction was set from 1960 to 2021 and any study that matched the search strategy and did not counterpose the exclusion criteria were revised.

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A total of 1,890 manuscripts were retrieved during the first step of the literature review process and only those which fulfil the inclusion criteria were included for the author's revision. After this process, a total of 221 abstracts were reviewed and those that had an exclusion criterion were removed. In the end, a total of 17 manuscripts were included in this review.

List of Databases and Information providers linked to Delphis

MEDLINE Complete
MEDLINE
Complementary Index
Academic Search Complete
Business Insights Global
ScienceDirect
Gale In Context: Global Issues
Journals@OVID
PASCAL Archive
Engineering Source
Gale In Context: Middle School
vLex
Regional Business News
Applied Science & Technology Source
Gale OneFile: CPI.Q
Business Source Complete
Computers & Applied Sciences Complete
JSTOR Journals
Education Source
SocINDEX with Full Text
FRANCIS Archive
Legal Source
Humanities Source
GreenFILE
Dentistry & Oral Sciences Source
Fuente Académica Premier
Library & Information Science Source
Library, Information Science & Technology Abstracts
eBook Index
Aerospace Research Central

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eBook Collection (EBSCOhost)

Art & Architecture Source

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Authors	Razdanet al. 1988	Annoml et.al 1990	Garcia, J 1994	Jaillard, 1995	Jha et. al 2002	Niaz et al. 2003	Mahajan et. al 2004	Faeh et.al 2009	Ezzati et.al 2011	Dhiman et. al 2018	Khattar et.al 2019	Lu et.al 2020	Burtscher et. al 2021	Ortiz et.al 2021	Liu et.al 2021
Did the review address a clearly focused question?	YES	YES	YES	YES	YES	YES	NO	YES	NO	YES	YES	YES	YES	YES	YES
Did the authors look for the right type of papers?	CT	NO	CT	CT	YES	CT	CT	YES	CT	YES	YES	YES	YES	YES	YES
Do you think all the important, relevant studies were included?	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable
Did the review's authors do enough to assess quality of the included studies?	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable
If the results of the review have been combined, was it reasonable to do so?	CT	YES	YES	NO	CT	CT	NO	CT	NO	CT	CT	CT	CT	CT	CT

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What are the overall results of the review?	YES	NO	YES	CT	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
How precise are the results?	NO	NO	NO	NO	YES	NO	NO	NO	NO	YES	YES	YES	YES	YES	YES
Can the results be applied to the local population ?	YES	NO	NO	YES	YES	YES	YES	YES	CT	YES	YES	YES	YES	YES	YES
Were all important outcomes considered ?	YES	CT	NO	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	YES	YES
Are the benefits worth the harms and costs?	CT	NO	CT	CT	YES	YES	CT	CT	CT	YES	YES	YES	YES	YES	YES

Supplementary Table 1 CASP analysis of the selected literature

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Reporting checklist for systematic review and meta-analysis.

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement

		Reporting Item	Page Number
Title			
	#1	Identify the report as a systematic review, meta-analysis, or both.	1
Abstract			
Structured summary	#2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	2
Introduction			
Rationale	#3	Describe the rationale for the review in the context of what is already known.	4

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Objectives	#4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
Methods			
Protocol and registration	#5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address) and, if available, provide registration information including the registration number.	4
Eligibility criteria	#6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rational	4
Information sources	#7	Describe all information sources in the search (e.g., databases with dates of coverage, contact with study authors to identify additional studies) and date last searched.	4
Search	#8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	#9	State the process for selecting studies (i.e., for screening, for determining eligibility, for inclusion in the systematic review, and, if applicable, for inclusion in the meta-analysis).	4
Data collection process	#10	Describe the method of data extraction from reports (e.g., piloted forms, independently by two reviewers) and any processes for obtaining and confirming data from investigators.	4
Data items	#11	List and define all variables for which data were sought (e.g., PICOS, funding sources), and any assumptions and simplifications made.	4
Risk of bias in individual studies	#12	Describe methods used for assessing risk of bias in individual studies (including specification of whether this was done at the study or outcome level, or both), and how this information is to be used in any data synthesis.	5
Summary measures	#13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Planned methods of analyis	#14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I2) for each meta-analysis.	5

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Risk of bias across studies	#15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	#16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
Results			
Study selection	#17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	#18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citation.	6
Risk of bias within studies	#19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see Item 12).	5
Results of individual studies	#20	For all outcomes considered (benefits and harms), present, for each study: (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot.	7-15
Synthesis of results	#21	Present the main results of the review. If meta-analyses are done, include for each, confidence intervals and measures of consistency.	7-15
Risk of bias across studies	#22	Present results of any assessment of risk of bias across studies (see Item 15).	7-15
Additional analysis	#23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7-15
Discussion			
Summary of Evidence	#24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., health care providers, users, and policy makers	16
Limitations	#25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	20
Conclusions	#26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	20

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Funding

Funding	#27	Describe sources of funding or other support (e.g., supply of data) for the systematic review; role of funders for the systematic review.	21

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This checklist was completed on 27. March 2021 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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BMJ Open

Chronic high altitude exposure and the epidemiology of ischemic stroke: A systematic review

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Primary Subject Heading:	Neurology
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Keywords:	Stroke < NEUROLOGY, ALTITUDE MEDICINE, EPIDEMIOLOGY

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Chronic high altitude exposure and the epidemiology of ischemic stroke: A systematic review

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Word count: 3,900

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Abstract

Introduction

About 5.7% of the world population resides above 1,500 m. It has been hypothesized that acute exposure to high-altitude locations can increase stroke risk, while chronic hypoxia can reduce stroke-related mortality.

Objective

This review aims to provide an overview of the available evidence on the association between long-term high altitude exposure and ischemic stroke.

Design

A systematic review was performed from January 1st, 1960, to December 1st, 2021, to assess the possible link between high altitude exposure and ischemic stroke. The AMED, EMBASE, Cochrane Library, PubMed, MEDLINE, the European PubMed Central and the Latin-American bibliographic database Scielo were accessed using the University of Southampton library tool Delphis. In this review we included population and individual based observational studies, including cross sectional and longitudinal studies except for those merely descriptive individual-based case reports. Studies were limited to humans living or visiting high altitude locations for at least 28 days as a cut-off point for chronic exposure.

Results

We reviewed a total of 1,890 abstracts retrieved during the first step of the literature review process. The authors acquired in full text as potentially relevant 204 studies. Only 17 documents met the inclusion criteria and were finally included. Ten studies clearly suggest that living at high altitudes may be associated with an increased risk of stroke, however, five studies suggest that altitude may act as a protective factor for the development of stroke, while two studies report ambiguous results.

Conclusions

This review suggest that the most robust studies are more likely to find that prolonged living at higher altitudes reduces the risk of developing stroke or dying from it. Increased irrigation due to angiogenesis and increased vascular perfusion might be the reason behind improved survival profiles among those living within this altitude range. In contrast, residing above 3,500 m, seems to be associated with an apparent

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increased risk of developing stroke, probably linked to the presence of polycythaemia and other associated factors such as increased blood viscosity.

Keywords: Stroke; High altitude; Hypoxia; Thrombosis; angiogenesis; Review

Strengths and limitations of this study

- This review analyzes the relationship between stroke and high altitude exposure using published data.
- We report a quantitative analysis of the risk factors associated with ischemic stroke at high altitude.
- One of the main limitations is the lack of average multi-country cohorts.
- Another limitation is that case reports were excluded from our analysis.

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Introduction

Stroke, also known as a cerebrovascular event, is a severe and often life-threatening medical condition that appears when the blood supply to part of the brain is interrupted [1]. It causes a rapid, acute, and potentially progressive brain function loss due to a disturbance in blood supply, oxygen, and nutrients that lasts at least 24 hours [2].

Cerebrovascular events or stroke is the second leading cause of death worldwide, affecting more than 16 million people each year[1]. Around 1 in 6 men and 1 in 5 women will have a stroke in their lifetime, and it is the third leading cause of disability worldwide [3,4]. Stroke affects people of all ages, though the causes of a stroke at a younger age are often different from those at older ages, especially in terms of risk factors and severity [5,6].

Clinical manifestations usually depend on the severity of the anoxia, the brain's location, the type and subtype of stroke, and the presence of other individual risk factors. The most common clinical manifestations are sudden unilateral weakness or numbness, diplopia, ataxia, and aphasia [1,7].

Stroke risk factors are often classified as traditional and non-traditional, and from this, they can be modifiable or non-modifiable [8]. In general terms, these risk factors have been classified as those which cause an ischemic stroke or haemorrhagic, being the most common and recognized as displayed in table 1.

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Table 1 Traditional modifiable and non-modifiable risk factors of stroke

Traditional risk factors			
		Non-modifiable	**Modifiable**
Ischemic Stroke	**Thrombotic**	Age Sex Ethnicity Socio-economic status Family History Arterial dissection	Waist circumference Alcohol misuse Obesity (BMI, WC, WHR) Diabetes Cigarette Smoking
	Embolic	Atrial fibrillation Intracardiac thrombus Heart valve disease Trauma and fractures Some types of Surgeries	Physical inactivity Apolipoprotein B to A1 Hyperlipidaemia***
	Systemic	Post-traumatic hypovolemia acute systemic hypoxia	
Haemorrhagic stroke	***ICH**	Vascular Malformations bleeding diatheses Trauma	Hypertension Cigarette smoking Obesity Waist -to-hip ratio
	****SAH**	Cocaine Abuse Amphetamines	Diet

*Intracerebral haemorrhage ** Subarachnoid haemorrhage *** High LDL serum levels

The risk of developing stroke increases with the presence of causal factors, which include arterial hypertension, atrial fibrillation, cigarette smoking, hyperlipidaemia, diabetes mellitus and acute infections such as COVID-19 [1,9]. Other modifiable factors are obesity, chronic kidney disease, excessive alcohol and cocaine consumption, sedentarism, psychological stress or depression [10–12]. Other factors such as Vitamin D or high-altitude may play a role but have been less studied than the traditional vascular risk factors [13–17].

Rationale for high altitude exposure and stroke

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Globally, at least 5.7% of the population lives above 1,500 meters, nevertheless, the association between high-altitude exposure and stroke has been poorly studied [14,15,18]. The available literature is based mainly on case reports due to acute exposure to extreme altitudes, rather than longer-term living among high-altitude dwellers [14,19–21]. The few reports available suggest that living in high-altitude regions (>2,500 m) increases the risk of developing thrombosis through hypoxia driven polycythaemia which leads to a hypercoagulation unbalance, that have been associated with increased risk for the development of atherothrombotic stroke [22–24].

Although most of the information comes from a case or case-series reports, no cohorts' studies have been published in this area. The only available information came from very few cross-sectional analysis that has found a significant association between living in high-altitude regions and having a greater risk of developing stroke, especially among younger populations [14,15,20,21].

Objective

To further explore this relationship, we have conducted a literature review of the available information in terms of the link between high-altitude chronic exposure and the risk of developing ischemic stroke.

Methods

Research Question

Does living at high-altitude increases the risk of developing ischemic stroke?

Study Design

A systematic review was conducted, including published scientific reports including original research, abstracts, editorial and commentary letters. We have used the preferred reporting guidelines for systematic reviews and meta-analyses (PRISMA). No protocol was registered for this review since PROSPERO does not currently accept registrations for scoping reviews, literature reviews or mapping reviews.

Data Source and search strategy

The literature review process was performed in English to cover the largest number of scientific databases and repositories containing academic literature from January 1st 1960 to December 1st 2021. The following libraries, repositories, and databases were accessed using the University of Southampton library tool

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Delphis: AMED, EMBASE, Cochrane Library, PubMed, MEDLINE, and Europe PMC. The Latin-American bibliographic database Scielo was also included in a separate search from its website repository.
The studies included in based on the following criteria:
Population: Adults from both sexes, all ages and ethnic background.

Study designs: All epidemiological analysis including observational and experimental full research, abstracts, editorials o commentary letters.

Exposure: We included every elevation above 2,000 m that was considered high altitude by the authors

Outcome of interest: Incidence and mortality rates

Inclusion criteria

This systematic literature review included all scientific papers that were written in English and reported epidemiological findings on the impact of stroke in different populations exposed to moderate to high altitudes. We set the cut-off point of short or long-term exposure in 28 days based on previously published data.
The main inclusion criteria were:
a) all manuscripts involving human subjects
b) all manuscripts that considered high altitude in the title or abstract and had the word stroke in the title.
c) all studies published between January 1st 1960 to December 1st 2021
The precise, full search strategy for all databases, registers, and websites, including any filters and limits used is described within the supplementary file 1.

Exclusion criteria

Exclusion criteria included all reports of thrombosis, deep vein thrombosis or venous thrombosis.
The list of the main exclusion criteria was:
a) Scientific papers concerning stroke that did not consider high altitude within the analysis.
b) This literature review excluded all in-vivo and in-vitro studies.
c) If the search term "stroke" referred to cardiac output or stroke volume, the results were excluded.
d) Case reports were excluded from the study.

Bias assessment

In order to reduce risk of any type of bias, the data extraction process was performed by EO and SC independently and in different times. To minimize errors while gathering information from any primary study, any disagreements was resolved after reaching consensus. In addition, while assessing the quality of primary studies, the CASP critical appraisal tool was used to reduce any tendency to search, interpret,

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favour, or recall any information previously adopted by any of the authors of this work (supplementary file 2).

Data Synthesis

In this section we completely reviewed all the manuscripts that met the inclusion criteria. With the information obtained, a quantitative analysis of the results using the CASP tool was performed. The information coming from each of the manuscripts was synthesized in two tables.

Patient and Public Involvement

No patient involved

Results

Literature review

Several anecdotal case reports and case series of people suffering from stroke-related disorders at high-altitude have been published since the late 1800s [25]. Despite this apparent relationship, very few well-conducted epidemiological studies have been conducted worldwide, and we have only included those studies that fulfilled our inclusion and exclusion criteria [14,15,20,21,26–36].

We included 1,890 manuscripts during the first step of the literature review process, and only those that fulfil the inclusion criteria were revised. After this process, the authors evaluated 204 abstracts. In the end, we included 17manuscripts within this literature review (Figure 1).

Figure 1 Prisma chart diagram for reporting literature review search and metanalysis

Specific studies

Study 1: The first study published by Razdan et al. in 1988 looked into the prevalence of stroke in elevations above 1,500 m [26]. They found that the crude prevalence of stroke was higher than the populations residing

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below this elevation (Table 2). The study was carried out in India and includes 91 cases of stroke, the sample included men and women between 5 and 65 olds [26].

Study 2: In 1990, Annoml et al. published an article about the incidence of cerebrovascular accidents in children with sickle cell disease residing at high and low altitudes in Saudi Arabia. They found that 4% and 5% of the 400 children included in the study who were living at low and high altitudes developed stroke. The authors did not explore any causal relationship or controlled for confounders. The results presented in that study are not significant, considering the total number (n=8) of children with sickle cell diseases and stroke [27].

Study 3: In 1994, Garcia-J reported the prevalence of risk factors of developing stroke at 2,500 m in a community in Peru; nevertheless, the study focused on the association between risk factors and stroke and not in the relationship with elevation [28]. In this study, the author discusses cerebrovascular events in people living above 2,500 m but does not analyse the role of hypoxia as a triggering or protective factor. We came to this work because it was conducted at high altitudes and had stroke as an outcome, nevertheless, no comments about hypoxia were given in this study.

Study 4: In 1995, Jaillard et al. published one of the most cited stroke documents at high-altitude. A document from Perú described the results from a "door to door" survey in Cuzco, a city located at 3,380 meters above sea level. In this study, men and women over 15 years of age were included and classified in -97% mestizos, 1.3% white, 1.1% Quechua, 0.1% Aymara and 0.2% were from other racial groups. They reported a crude prevalence rate of 6.47 per 1,000 population; nevertheless, comparisons were performed using standard populations from other locations, while not giving enough information about the association of high-altitude and stroke per se. This study's evidence is limited since the comparison was performed with other reports at lower elevations; therefore, the conclusions should be taken cautiously [20].

Study 5: A Population-based analysis published by Jha et al. in 2002 reported a significantly higher incidence of stroke among patients admitted to a single tertiary hospital in India. They classified those patients that came from lower altitudes (<3,000 m) and higher altitudes (>3,000 m). They reported that the majority of stroke cases at high-altitude were presented in otherwise-healthy young men (<45 years old) exposed to high-altitude for months (nine months on average). The hospital admission rate at high-altitude was reported to be 12 times greater (12.8/1,000) than at lower altitudes (1.0/1,000), being this results statistically significant [14].

Study 6 In 2003, Niaz et al. conducted a case-control study to describe the prevalence of stroke above 4,571 m in India . They included 4,000 soldiers (20-40 y/o) from a military hospital to observe stroke occurrence. They reported 10 cases of stroke among young men patients from the high-altitude group

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versus only one case from the lower land cohort. Although no further information about size effect or standardization of the sample was provided, the researchers concluded that living at elevations above 4,500 m was associated with ten times more risk of developing stroke. The authors concluded that chronic (> 28 days) hypobaric hypoxia exposure it is associated with a higher risk of developing stroke, being massive ischemic stroke the most frequent presentation [15].

Study 7 In 2004, Mahajan et al. conducted a cross-sectional study in Himachal, Pradesh, India, among 100 patients between men and women some had risk factors such as hypertension or smoked from urban and rural areas hospitalized in a tertiary level hospital located at an altitude of 2200 meters. They reported that the incidence of stroke was lower when compared with previously published studies from India. Although information from the type of stroke and clinical features are available, the role of elevation was not discussed in depth. The authors compared the results with other studies, but no confounding analysis was made for the current report [29].

Study 8: An ecological study published by Faeh et al. in 2009 reported that stroke mortality is lower in higher locations in Switzerland (Faeh D et al., 2009). They reported that mortality from stroke decrease 12% per every 1000 m gained in elevation. The effect of altitude on the cardiovascular and cerebrovascular disease was assessed using sociodemographic information, places of birth and residence among men and women between 40 and 84 years of age living at altitudes of 259 to 1960 m. The protective effect of living at higher altitudes on coronary heart disease and stroke mortality was consistent and became stronger after adjustment for potential confounders [21].

Study 9: An ecological presented by Ezzati et al. in 2011, used from the National Elevation Dataset, National Centre for Health Statistics, and US Census. The authors analyzed the crude association of mean county altitude with life expectancy and mortality from ischaemic heart disease (IHD), stroke, chronic obstructive pulmonary disease (COPD), and cancers. Living at higher altitudes showed a beneficial association with ischemic heart disease mortality, but the results were not statistically significant for stroke [30].

Study 10: A prospectively collected Study in India published by Dhiman et al. 2018 reported the evolving Pattern and Outcome of Stroke at Moderate Altitude. This prospective, observational study was carried out at 2,000 meters above sea level and included the clinical features and risk factors profile of 235 patients diagnosed with a stroke between men and women with an average age of 62 years. The results only compared how the data compared with a report published 15 years ago, and the authors concluded that the occurrence of stroke decreased among hospitalized patients compared to the 2004 report [29,31].

Study 11: An Ecological study design in Nepal published by Khattar et al. in 2019 demonstrated that ischemic stroke is most likely to occur at elevations above 3,000 m and although this diagnosis is not

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technically considered stroke, the physiopathology of it might indicate some relationship with the development of ischemic cerebrovascular events. Despite some of these findings that targeted high-altitude as a possible risk factor for developing stroke, in 2009 [32] (Table 2). The study included 21 patients with stroke, 76% were men with a mean average age of 56 years, medical comorbidities included hypertension and diabetes were reported.

Study 12: An Hospital-based comparative study in Tibet and Beijing with 236 and 1021 in patients with acute ischemic stroke published by Lu et al in 2020 demonstrated that patients diagnosed with acute ischemic stroke located at 3650 m (Tibet) above sea level were younger than those who lived 40 m (Beijing) above sea level ($P < 0.001$); also, in Tibet, there was a higher proportion of cerebrovascular accidents in young adults, which may be associated with erythrocytosis [33].

Study 13: An ecological study published by Burtscher et al in 2021 carried out in Austria demonstrated that mortality for all causes including stroke decrease as height increases, demonstrating a linear association; the mortality rate was lower around 15-20% in those who live at altitudes ranging from 1000 to 2000 m. It is important to note that the causes of death most affected by height are cardiovascular diseases and cancer, since at higher altitudes people have a lower mortality rate from these diseases[34].

Study 14: An ecological analysis of all stroke hospital admissions, mortality rates, and disability-adjusted life years in Ecuador published by Ortiz et al in 2021 demonstrated that high altitude populations had lower stroke mortality in men [OR: 0.91 (0.88–0.95)] and women [OR: 0.83 (0.79–0.86) and had a significant lower risk of getting admitted to the hospital when compared with the low altitude group in men [OR: 0.55 (CI 95% 0.54–0.56)] and women [OR: 0.65 (CI 95% 0.64–0.66)][35]

Study 15: Ecological Study carried out in China published by Liu et at demonstrated that Ischemic strokes at high altitudes were characterized by younger ages and larger infract volumes, also risk factors such as atherosclerosis, diabetes mellitus, coronary disease, and hyperlipidaemia were significantly lower than those in high-altitude areas [36].

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Table 2 Critical appraisal and summary of the literature.

Author and year	Tittle	Sample, setting and elevation	Aim	Design, type and length of Exposure	Main findings
Razdan et al. 1988	Cerebro-vascular Disease in Rural Kashmir, India	- The study was performed in Kuthar Valley, India -Elevation 1,530 m.	To find the prevalence and trends related to stroke in this region	-A cross sectional survey to find if population has suffered from stroke.	- 91 cases of stroke were detected - Crude prevalence rate 143/100,000 - Proportion in men (69,23%) and women (30,77%)
Annoml et.al 1990	Cerebrovascular accidents in children with sickle cell disease residing at high and low altitudes of Saudi Arabia	- The study was performed in Saudi Arabia -8 stroke cases -3,000 m	- To compare the clinical and radiological indications of stroke in sickle cell patients constantly residing at low and high-altitudes.	-Analyses of 8 cases of stroke among children with sickle cell disease -extracted from 400 cases detected in Saudi Arabia.	-The incidence of stroke in children with sickle cell disease similar regardless elevation.
Garcia, J 1994	Cerebrovascular Disease: Risk Factors analysis of Natives living at high-altitude.	-The study was performed in Peru -50 patients with a confirmed diagnosis of stroke and 52 healthy patients -2,500 m	To understand the role of different risk factors in patients located at high altitudes	Retrospective case-control study - Population selection based on criteria (cases), random population (controls)	- Atherosclerosis was the most important risk factor for CVD in the highland natives.
Jaillard, 1995	Prevalence of Stroke at High-altitude (3380 m) in Cuzco, a Town of Peru A Population-Based Study	- The study was performed in Peru -3,246 subjects interviewed -3,800 m	to study the prevalence of stroke and its association with stroke risk factors in a high-altitude location.	A door to door population-based analysis using a survey as a tool for collecting data.	-Age, polycythemia, high consumption of alcohol, and area of residence were associated with higher stroke prevalence.
Jha et. al 2002	Stroke at High-altitude: Indian Experience	- The study was performed in India -4,270 m	-To determine the relationship between stroke and high-altitude.	-Information was prospectively collected from the Command Hospital in India	-Long-term stay at high-altitude was associated with higher risk of developing stroke.
Niaz et al. 2003	Stroke at high altitude: Indian experience	-The study was performed in Chandimandir, India -Elevation: 4,200 m.	-To identify the presence of stroke using different clinical, imaging and laboratory assessments.	-A cohort analysis of stroke prevalence and mortality from 1998-2000.	-Stroke hospital admission rates was higher among high altitude residents (13.7/1000) versus low altitude (1.05/1000)
Mahajan et. al 2004	Stroke at moderate altitude	-this study was performed in Himachal, Pradesh in the Sub-Himalayan ranges. -elevation: 2,200 m	To understand the clinical profile, presence of various risk factors for stroke at moderate altitude.	-A cross sectional analysis of stroke admission in a tertiary hospital from India	-Patients aged up to 45 years were defined as stroke in young. -Complete clinical, radiological and neurological examination was performed.
Faeh et.al 2009	Lower Mortality from Coronary Heart Disease and Stroke at Higher Altitudes in Switzerland.	-The study was performed in Switzerland. - 1 641 144 Swiss and foreign nationals born in Switzerland -259 to 1960 m.	Examine mortality from coronary heart disease (CHD) and stroke in its association with high-altitude living.	-A longitudinal analysis of mortality data from 1990 to 2000.	-Living at higher altitude was associated with less CHD and less stroke mortality.
Ezzati et.al 2011	Altitude, life expectancy and mortality from ischaemic heart disease, stroke, COPD and cancers: national population-based analysis of US counties	The Study was performed in USA -Mortality and population data for 2001 through 2005. -12.1 million death records -500-1000 m -1000- 1500 m -1500 > m	Inspect the relationship of life expectancy and mortality from particular diseases in relationship to altitude.	-Ecological study of the association of mean county altitude with life expectancy and mortality from IHD, Stroke, COPD and cancer	- Counties above 1500 m had longer life expectances than those within 100m of sea level by 1.2 - 3.6 years for men and 0.5 -2.5 years for women.

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Dhiman et. al 2018	The Evolving Pattern and Outcome of Stroke at Moderate Altitude	- The study was performed in India -Men and women with an average age of 62 years Sub-Himalayan region of India. -235 patients with stroke were consecutively admitted into a tertiary hospital. -2,000 m	The aim was to compare the clinical profile, risk factors, and outcome in hospitalized patients of stroke in a Tertiary Care Hospital situated at moderate altitude	-A prospectively collected Study in India. -A comparative analysis was performed with a previous study performed 15 years prior of the current study.	-Ischemic stroke was noted in 74%, and 26% had haemorrhagic stroke (HS). -Male accounted 58% of the cases and female for 42%. -Overall HS had poorer outcome. -There occurrence of stroke has decreased among hospitalized patients at moderate altitude.
Khattar et.al 2019	Cerebral Venous Thrombosis at High-altitude: A Retrospective Cohort of Twenty-one Consecutive Patients	-The study cohort comprised 21 patients in Nepal. CVT was the diagnosis of interest. -elevation 3,048 m	Investigate the characteristics and treatment outcomes of patients who suffered CVT at high-altitude in eastern Nepal.	- Retrospectively reviewed all patients presenting with clinical and radiographic evidence of cerebral venous sinus thrombosis treated.	21 patients of which 76% were men with an average age of 56. -men were found to have a higher risk for CVT at high-altitude -All patients presented with evidence of haemorrhagic conversion on the initial brain CT.
Lu et.al 2020	Characteristics of acute ischemic stroke in hospitalized patients in Tibet: a retrospective comparative study	- The study was performed in Tibet and Beijing - 236 and 1021 inpatients with acute ischemic stroke from People's Hospital of Tibet Autonomous Region and Peking University First Hospital -40 m and 3650 m above sea level	This study aimed to analyse the clinical characteristics of AIS patients at high-altitude regions through a hospital-based comparative study between Tibet and Beijing.	-Hospital-based comparative study	- In Tibet, AIS patients were relatively younger, and anterior circulation infarctions were more common - Young adult stroke, erythrocytosis, and hyperhomocysteinemia were more frequent in PHOTAR patients
Burtscher et. al 2021	Does living at moderate altitudes in Austria affect mortality rates of various causes? An ecological study	- The study was performed in Austria - 467 834 deaths -Elevations: <251 m, 251–500 m, 501–750 m, , 751–1000, 1001-2000 m	This study tested the hypothesis that living at moderate altitudes (up to 2000 m) is associated with reduced mortality from all causes.	An ecological study	- ASMRs for residents living in higher (>1000 m) versus lower (<251 m) altitude regions (with agriculture employment below 3%) were 485.8 versus 597.0 (rate ratio and 95% CI 0.81 (0.72 to 0.92); p<0.001) for men and 284.6 versus 365.5 (0.78; 0.66 to 0.91); p=0.002) for women.
Ortiz et.al 2021	Stroke-Related Mortality at Different Altitudes: A 17-Year Nationwide Population-Based Analysis From Ecuador	- The study was performed in Ecuador - A total of 38,201 deaths and 75,893 hospital admissions due to stroke -Elevations low altitude (<1,500 m), moderate altitude (1,500–2,500 m), high altitude (2,500–3,500 m) and very high altitude (3,500–5,500 m)	Elucidate the association between stroke and altitude using four different elevation ranges	-An ecological analysis of all stroke hospital admissions, mortality rates, and disability-adjusted life years in Ecuador was performed from 2001 to 2017	- A total of 38,201 deaths and 75,893 stroke-related hospital admissions were reported -High altitude populations (HAP) had lower stroke mortality in men [OR: 0.91 (0.88–0.95)] and women [OR: 0.83 (0.79–0.86)].
Liu et.al 2021	Acute Ischemic Stroke at High Altitudes in China: Early Onset and Severe Manifestations	-The study was performed in China -892 cases of first-ever acute ischemic strokes at altitudes of 20, 2550, and 4200 m in China (697 cases from Penglai, 122 cases from Huzhu, and 73 cases from Yushu).	Describe and investigate the characteristics of acute ischemic stroke at different altitudes in China	Ecological Study	-Ischemic strokes at high altitudes were characterized by younger ages (69.14 ± 11.10 vs. 64.44 ± 11.50 vs. 64.45 ± 14.03, p < 0.001) and larger infract volumes (8436.37 ± 29,615.07 mm3 vs. 17,213.16 ± 47,044.74 mm3 vs. 42,459 ± 84,529.83 mm3, p < 0.001

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Discussion

The results from this review suggest that stroke seems to be more likely to occur in very high-altitude locations (>3,500 m) when the exposure is longer than 28 days, especially among younger people (< 45 y/o). On the other hand, when people live above 1,500 m and below 3,500 seems like there is a protective effect for stroke, probably triggered by better adaptation to hypoxia, efficient enough to reduce the likelihood of dying when compared to lower altitudes; nevertheless, no information is available about the exact point in which this protective effect become a risk factor.

It has been challenging to define how high altitude exposure can be defined and where the threshold is located in terms of mild or severe hypoxia [37]. For instance, Imray in 2017uses a classification of high-altitude exposure according to the recommendations from the International Society of Mountain Medicine, a categorization that seems to be the most pragmatic [38]. The author defined low altitude everything located below 1,500 m, moderate or intermediate altitude between 1,500 to 2,500 m, high-altitude from 2,500 to 3,500 m, the very high-altitude from 3,500 m to 5,800, more than 5,800 extreme high-altitudes and above the 8,000 m is considered the death zone [38].

As humans acclimatize to high-altitude, adverse and often mild secondary effects can occur in response to hypoxia. Some of these adverse effects are linked to the increased blood viscosity due to polycythemia, augmented pulmonary arterial pressure, and sometimes, they are linked to a proposed hypercoagulation unbalance [22,24].

These consequences might be increasing the risk of forming an atherothrombotic plaque resulting in a stroke or myocardial infarction (MI) or venous-thrombotic events, resulting in deep venous thrombosis (DVT) or pulmonary embolism (PE) [39–41]. Although information about the time of exposure is scarce, the longer the exposure, the higher the risk [21,42].

Acute exposure to hypobaric hypoxia produces several compensatory physiological effects that can last hours, days, months, or years. The essential mechanisms are: increasing the heart and respiratory rates, a secondary polycythemia, haemoconcentration derived from reduced plasma volume caused by respiratory evaporative water loss and polyuria, and increased ventilatory response [37,43,44]. When acute exposure last longer than 28 days, more efficient and prolonged mechanisms take place, including sustained polycythemia, endothelium changes, reduced vascular resistance, nitric oxide-mediated hypotension, and angiogenesis [45–48]. Acute exposure to high-altitude hypoxia triggers a series of events that produce a hypercoagulable state [24]. This hypercoagulability state is boosted by dehydration, haemoconcentration, and polycythemia. When combined with dehydration (due to tachyphemia and extenuate physical activity) and limited mobilization (sleeping in tents and secluded spaces), these factors produce the perfect scenario for increasing vascular stasis and thrombosis [22,37,49].

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When humans are exposed continuously to hypoxia, they develop adaptative mechanisms that are far more efficient than those observed in newcomers [50–53]. These last-longing mechanisms include anatomical (wider chests, shorter and lighter bodies, etc.), embryological (smaller fetus and placentas), circulatory (improved maximum flow output and higher pulmonary arterial pressure), and respiratory adaptations (improved hypoxic ventilatory response and oxygen diffusion capacities) [52,54–56]. Chronic exposure to hypobaric hypoxia leads to the development of more subtle compensatory mechanisms. These factors include long-term erythrocytosis, angiogenesis, capillary remodelling, and an improved ventilatory response [57–60] (Figure 2).

Once the general context of acute or chronic hypobaric hypoxia has been described, the main intrigue is which elevation is enough to generate compensatory mechanisms capable of reducing the risk of developing stroke and when these mechanisms become detrimental. After reviewing the current literature, the information available suggests that a window around 2,000 to 3,500 meters of elevation might be enough to generate some protective mechanisms (i.e., angiogenesis or vascular remodelling) against stroke [21,45,48].

In elevations below 2,000 meters, the degree of compensation might not be enough to ensure a protective effect, while above 3,500 meters, the adaptative compensatory mechanisms such as significant polycythemia and vascular stasis might increase the risk of thrombosis and, therefore, the risk of developing stroke [14,22,32] (Figure 2).

Figure 2 Proposed mechanisms and hypothesized physiopathology at low altitude (< 2,500 m), at high altitude (2,500-3,500 m) and very high altitude (>3,500 m). BP: Barometric pressure and O_2 is the oxygen availability in relationship to sea level FiO_2. : Angiogenesis occurs at different elevations above sea level but during the hypoxic beneficial window, polycythemia and red blood cell as well as platelet adhesiveness is not significant as above 3,500 m, thus, the protective effect reaches its maximum. Above 3,500 m although angiogenesis is present, the significantly high hematocrit and polycythemia increase the risk of blood stasis and thrombogenesis.

The information is still contradictory and opposed from one study to another. The few studies available have many limitations, and confounders' control was low in most of them. Nevertheless, very few studies that are better controlled and designed support some of our statements above. This report was designed to guide clinicians and researchers who are currently working with stroke and wanted to understand the role of

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elevation and hypobaric hypoxia for developing stroke while we suggest that further analysis and well-controlled studies are needed.

Limitations

Several limitations were found, including scarce information, conflicting results, and data lack when adjusting for confounders. In this sense, more research is needed to obtain a definitive answer; nevertheless, the information provided in this document can be used as an updated guide of the possible role of high-altitude exposure as a risk factor for developing a stroke.

Conclusions

This review suggest that the most robust studies tend to advocate that prolonged living at higher altitudes reduces the risk of developing stroke or dying from it. Increased irrigation due to angiogenesis and increased vascular perfusion might be the reason behind improved survival profiles among those living within this range. In contrast, residing high altitude locations, especially above 3,500 m, is associated with an apparent increased risk, probably linked to the presence of polycythaemia and other factors such as increased blood viscosity and the presence of a proposed hypercoagulable state might increase the risk of developing stroke among those exposed to very high altitudes. It seems clear that short-term exposures to very high altitudes are a risk factor for developing a stroke. The available scientific literature suggests that above 3,500 to 4,000, the risk of developing stroke increases, especially if the exposure is acute among non-adapted populations.

It is important to note that one of the main limitations presented by some of the studies analysed was the lack of analysis of risk factors related to stroke, in addition, the level of education, socioeconomic level, or living conditions of the participants were not analysed. We also highlight that certain risk factors such as diabetes, arteriosclerosis, coronary heart disease, or hyperlipidaemia have a lower prevalence in people living in high altitude areas

Recommendations

Further investigations are needed to explore the role of socioeconomic variables and traditional risk factors among populations located at different elevations. Understanding the cultural and social differences between highlanders and lowlanders will permit more suitable and robust conclusions when elevation becomes a protective factor and when the parabola from lower risk to higher risk shifts above a certain altitude.

Abbreviations

ABP: Arterial Blood pressure

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BMI: Body max index
BP: Barometric Pressure
COPD: Chronic obstructive pulmonary disease
CVD: Cerebrovascular disease
CVT: Cerebral venous thrombosis
CT: Computed tomography
HS: haemorrhagic stroke
IHD: Ischemic heart diseases
INEC: Ecuadorian Institute of Statistics and Census
IS: Ischemic stroke
ICD-10: 10th revision of the International Statistical Classification of Diseases and Related Health Problems
I60: Nontraumatic subarachnoid haemorrhage
I61: Nontraumatic intracerebral haemorrhage
I62: Cerebral infarction
I64: Other and unspecified nontraumatic intracranial haemorrhage
MRI: Magnetic Resonance Imaging
SES: Socioeconomic status

Availability of supporting data

All the information used for this analysis can be found online throughout the several medical databases available

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

EOP, SPC, EV, GV and PR contributed with the review of the bibliography and preparation of the manuscript, EV and EOP oversaw the entire revision of the document and finalizing the version end of our manuscript

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Ethics Statement

This secondary data analysis of publicly available anonymized data received ethical approval from the University of Southampton with the Faculty of Medicine Ethics Committee ERGO 51422.R3 number.

References

For peer review only

1 Hankey GJ. Stroke. *The Lancet* 2017;**389**:641–54. doi:10.1016/S0140-6736(16)30962-X

2 Wilson D, Charidimou A, Ambler G, *et al.* Recurrent stroke risk and cerebral microbleed burden in ischemic stroke and TIA: a meta-analysis. *Neurology* 2016;**87**:1501–10.

3 Seshadri S, Wolf PA. Lifetime risk of stroke and dementia: current concepts, and estimates from the Framingham Study. *Lancet Neurol* 2007;**6**:1106–14.

4 Johnson W, Onuma O, Owolabi M, *et al.* Stroke: a global response is needed. *Bull World Health Organ* 2016;**94**:634.

5 Fullerton HJ, Wu YW, Zhao S, *et al.* Risk of stroke in children: ethnic and gender disparities. *Neurology* 2003;**61**:189–94.

6 deVeber GA, Kirton A, Booth FA, *et al.* Epidemiology and outcomes of arterial ischemic stroke in children: the Canadian Pediatric Ischemic Stroke Registry. *Pediatr Neurol* 2017;**69**:58–70.

7 Coleman ER, Moudgal R, Lang K, *et al.* Early rehabilitation after stroke: a narrative review. *Curr Atheroscler Rep* 2017;**19**:1–12.

8 Bridgwood B, Lager KE, Mistri AK, *et al.* Interventions for improving modifiable risk factor control in the secondary prevention of stroke. *Cochrane Database Syst Rev* 2018.

9 Nannoni S, de Groot R, Bell S, *et al.* Stroke in COVID-19: a systematic review and meta-analysis. *Int J Stroke* 2021;**16**:137–49.

10 Cheng Y-C, Ryan KA, Qadwai SA, *et al.* Cocaine use and risk of ischemic stroke in young adults. *Stroke* 2016;**47**:918–22.

11 Everson SA, Roberts RE, Goldberg DE, *et al.* Depressive symptoms and increased risk of stroke mortality over a 29-year period. *Arch Intern Med* 1998;**158**:1133–8.

12 Guzik A, Bushnell C. Stroke Epidemiology and Risk Factor Management. *Contin Lifelong Learn Neurol* 2017;**23**:15–39.

13 Szawarski P(1), Tam EWY(2), Richards P(3). Stroke at high altitude. *Hong Kong Med J* 2012;**18**:261.

14 Jha SK, Anand AC, Sharma V, *et al.* Stroke at high altitude: Indian experience. *High Alt Med Biol* 2002;**3**:21–7.

15 Niaz A, Nayyar S. Cerebrovascular stroke at high altitude. *J Coll Physicians Surg--Pak JCPSP* 2003;**13**:446–8.

16 Pilz S, Dobnig H, Fischer JE, *et al.* Low vitamin D levels predict stroke in patients referred to coronary angiography. *Stroke* 2008;**39**:2611–3.

17 Gürdal A, Keskin K, Orken DN, *et al.* Evaluation of Epicardial Fat Thickness in Young Patients With Embolic Stroke of Undetermined Source. *The neurologist* 2018;**23**:113–7.

18 Burtscher M. Effects of Living at Higher Altitudes on Mortality: A Narrative Review. *Aging Dis* 2014;**5**:274.

For peer review only

19 Chan T(1), Mak HKF(1), Wong WWY(2), *et al.* Acute ischaemic stroke during short-term travel to high altitude. *Hong Kong Med J* 2012;**18**:63–5.

20 Jaillard AS, Hommel M, Mazetti P. Prevalence of stroke at high altitude (3380 m) in Cuzco, a town of Peru. *Stroke* 1995;**26**:562–8.

21 Faeh D, Gutzwiller F, Bopp M. Lower mortality from coronary heart disease and stroke at higher altitudes in Switzerland. *Circulation* 2009;**120**:495–501. doi:10.1161/CIRCULATIONAHA.108.819250

22 Zangari M, Fink L, Tolomelli G, *et al.* Could hypoxia increase the prevalence of thrombotic complications in polycythemia vera? *BLOOD Coagul FIBRINOLYSIS* 2013;**24**:311–6. doi:10.1097/MBC.0b013e32835bfdb9

23 Gupta N, Ashraf MZ. Exposure to High Altitude: A Risk Factor for Venous Thromboembolism? *Semin Thromb Hemost* 2012;**38**:156–63. doi:10.1055/s-0032-1301413

24 Kotwal J, Apte CV, Kotwal A, *et al.* High altitude: A hypercoagulable state: Results of a prospective cohort study. *Thromb Res* 2007;**120**:391–7. doi:10.1016/j.thromres.2006.09.013

25 George J. The central asian expedition of capt. Roberovsky and lt. Kkozloff. 1896.

26 Razdan S, Koul RL, Motta A, *et al.* Cerebrovascular disease in rural Kashmir, India. *Stroke* 1989;**20**:1691–3.

27 Annobil SH, Omojola MF, Adzaku FK, *et al.* Cerebrovascular accidents (strokes) in children with sickle cell disease residing at high and low altitudes of Saudi Aurabia. *Ann Trop Paediatr* 1990;**10**:191–8.

28 García J. Enfermedad cerebrovascular: factores de riesgo en los adultos nativos de la altura, HCRI-IPSS, 1989-1992. *Rev Peru EpidemiolOnline* 1994;**7**:67–72.

29 Mahajan SK, Kashyap R, Sood BR, *et al.* Stroke at moderate altitude. *JAPI* 2004;**52**:699–702.

30 Ezzati M, Horwitz ME, Thomas DS, *et al.* Altitude, life expectancy and mortality from ischaemic heart disease, stroke, COPD and cancers: national population-based analysis of US counties. *J Epidemiol Community Health* 2012;**66**:e17–e17.

31 Dhiman D, Mahajan SK, Sharma S, *et al.* The evolving pattern and outcome of stroke at moderate altitude. *J Neurosci Rural Pract* 2018;**9**:68.

32 Khattar NK, Sumardi F, Zemmar A, *et al.* Cerebral Venous Thrombosis at High Altitude: A Retrospective Cohort of Twenty-one Consecutive Patients. 2019.

33 Lu Y, Zhuoga C, Jin H, *et al.* Characteristics of acute ischemic stroke in hospitalized patients in Tibet: a retrospective comparative study. *BMC Neurol* 2020;**20**:380. doi:10.1186/s12883-020-01957-0

34 Burtscher J, Millet GP, Burtscher M. Does living at moderate altitudes in Austria affect mortality rates of various causes? An ecological study. *BMJ Open* 2021;**11**:e048520. doi:10.1136/bmjopen-2020-048520

35 Ortiz-Prado E, Espinosa PS, Borrero A, *et al.* Stroke-Related Mortality at Different Altitudes: A 17-Year Nationwide Population-Based Analysis From Ecuador. *Front Physiol* 2021;**12**:1680. doi:10.3389/fphys.2021.733928

36 Liu M, Yan M, Guo Y, *et al.* Acute Ischemic Stroke at High Altitudes in China: Early Onset and Severe Manifestations. *Cells* 2021;**10**:809. doi:10.3390/cells10040809

For peer review only

37 West JB, Schoene RB, Milledge JS, *et al.* High altitude medicine and physiology. Hodder Arnold London 2007. http://www.jrnms.com/wp-content/uploads/2014/05/JRNMS-95-1-40-43.pdf (accessed 16 Aug 2016).

38 Imray C, Booth A, Wright A, *et al.* Acute altitude illnesses. *Bmj* 2011;**343**:d4943.

39 Brill A, Suidan GL, Wagner DD. Hypoxia, such as encountered at high altitude, promotes deep vein thrombosis in mice. *J Thromb Haemost JTH* 2013;**11**:1773–5. doi:10.1111/jth.12310

40 Khalil KF, Saeed W. Pulmonary Embolism in Soldiers Serving At High Altitude. *JCPSP-J Coll PHYSICIANS Surg Pak* 2010;**20**:468–71.

41 Kumar S(1 2). High altitude induced deep venous thrombosis: A study of 28 cases. *Indian J Surg* 2006;**68**:84–8.

42 Zavanone C, Panebianco M, Yger M, *et al.* Cerebral venous thrombosis at high altitude: A systematic review. *Rev Neurol (Paris)* 2017;**173**:189–93. doi:10.1016/j.neurol.2016.11.004

43 Azad P, Haddad GG. Molecular Basis of Hypoxia-Induced Excessive Erythrocytosis of High Altitude. In: *A74. RESPIRATORY HEALTH: ALTITUDE, ENVIRONMENTAL EXPOSURES, AND REHABILITATION*. American Thoracic Society 2018. A2366–A2366.

44 Steinback CD, Poulin MJ. Ventilatory responses to isocapnic and poikilocapnic hypoxia in humans. *Respir Physiol Neurobiol* 2007;**155**:104–13.

45 Ortiz-Prado E, Natah S, Srinivasan S, *et al.* A method for measuring brain partial pressure of oxygen in unanesthetized unrestrained subjects: the effect of acute and chronic hypoxia on brain tissue PO(2). *J Neurosci Methods* 2010;**193**:217–25. doi:10.1016/j.jneumeth.2010.08.019

46 Blitzer ML, Loh E, Roddy M-A, *et al.* Endothelium-derived nitric oxide regulates systemic and pulmonary vascular resistance during acute hypoxia in humans. *J Am Coll Cardiol* 1996;**28**:591–6.

47 Stenmark KR, Pugliese SC, Poth J, *et al.* The Effects of Chronic Hypoxia on Inflammation and Pulmonary Vascular Function. In: *Pulmonary Hypertension*. Springer 2016. 83–103.

48 Dunn JF, Wu Y, Zhao Z, *et al.* Training the Brain to Survive Stroke. *PLoS ONE* 2012;**7**:e45108. doi:10.1371/journal.pone.0045108

49 Bhalla IP, Naqui S, Ohri VC, *et al.* Lipid profile in polycythaemia of high altitude. *J Assoc Physicians India* 1988;**36**:374–5.

50 Brutsaert T. Why are high altitude natives so strong at high altitude? Nature vs. nurture: genetic factors vs. growth and development. In: *Hypoxia*. Springer 2016. 101–12.

51 Kollias J, Buskirk ER, Akers RF, *et al.* Work capacity of long-time residents and newcomers to altitude. *J Appl Physiol* 1968;**24**:792–9.

52 Moore LG. Measuring high-altitude adaptation. *J Appl Physiol* 2017;**123**:1371–85.

53 Vincent J, Hellot MF, Vargas E, *et al.* Pulmonary gas exchange, diffusing capacity in natives and newcomers at high altitude. *Respir Physiol* 1978;**34**:219–31.

54 Beall CM, Brittenham GM, Strohl KP, *et al.* Hemoglobin concentration of high-altitude Tibetans and Bolivian Aymara. *Am J Phys Anthropol* 1998;**106**:385–400.

For peer review only

55 Gassmann NN, van Elteren HA, Goos TG, *et al.* Pregnancy at high altitude in the Andes leads to increased total vessel density in healthy newborns. *J Appl Physiol* 2016;**121**:709–15.

56 Huicho L, Cabala-Peralta P, van-Elteren H-A, *et al.* Pregnancy at high altitude in the Andes leads to increased total vessel density in healthy newborns. 2016.

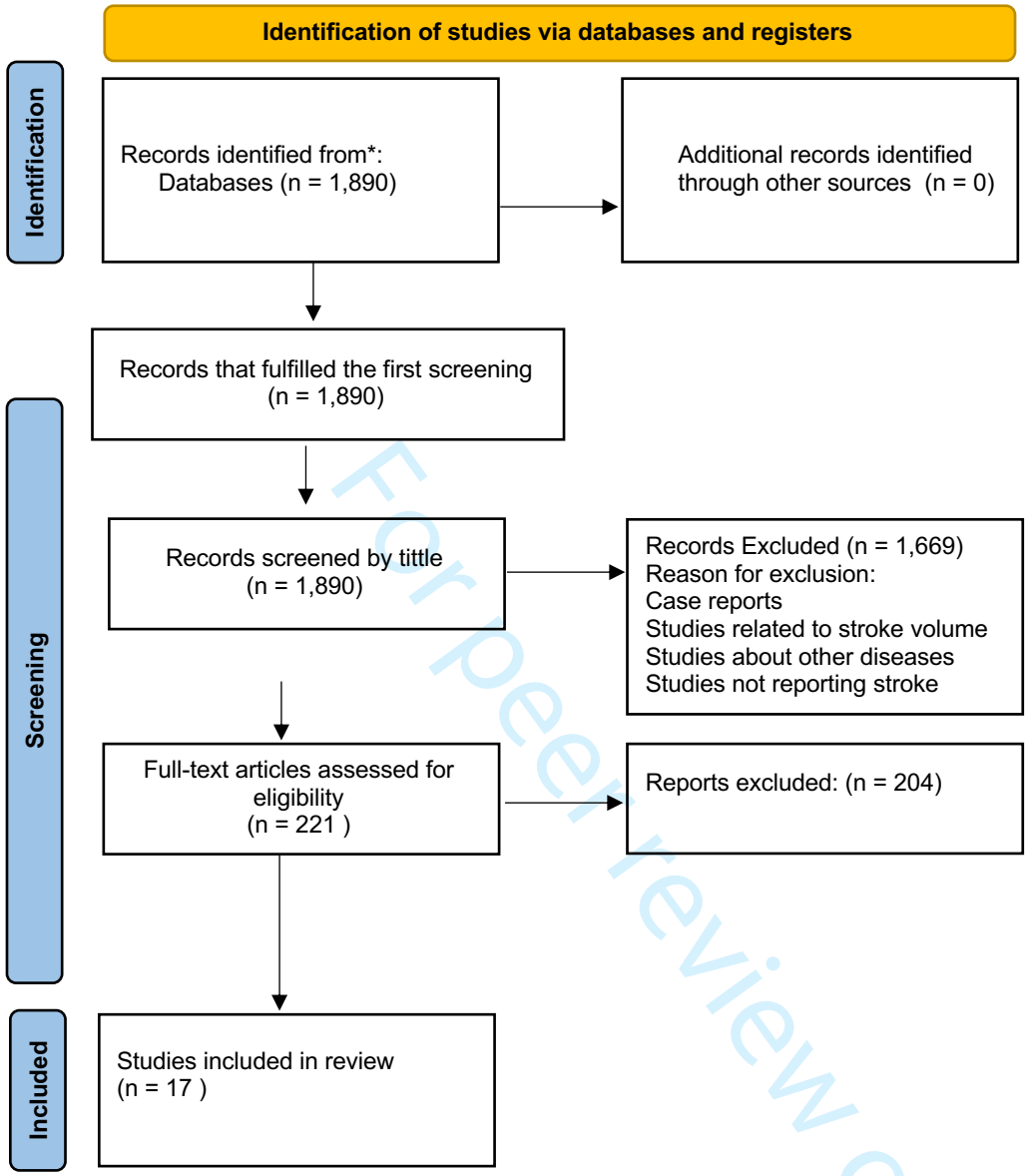
57 Chiodi H. Respiratory adaptations to chronic high altitude hypoxia. *J Appl Physiol* 1957;**10**:81–7.

58 León-Velarde F, Richalet J-P. Respiratory control in residents at high altitude: physiology and pathophysiology. *High Alt Med Biol* 2006;**7**:125–37.

59 Naeije R. Physiological adaptation of the cardiovascular system to high altitude. *Prog Cardiovasc Dis* 2010;**52**:456–66.

60 Ostadal B, Kolar F. Cardiac adaptation to chronic high-altitude hypoxia: beneficial and adverse effects. *Respir Physiol Neurobiol* 2007;**158**:224–36.

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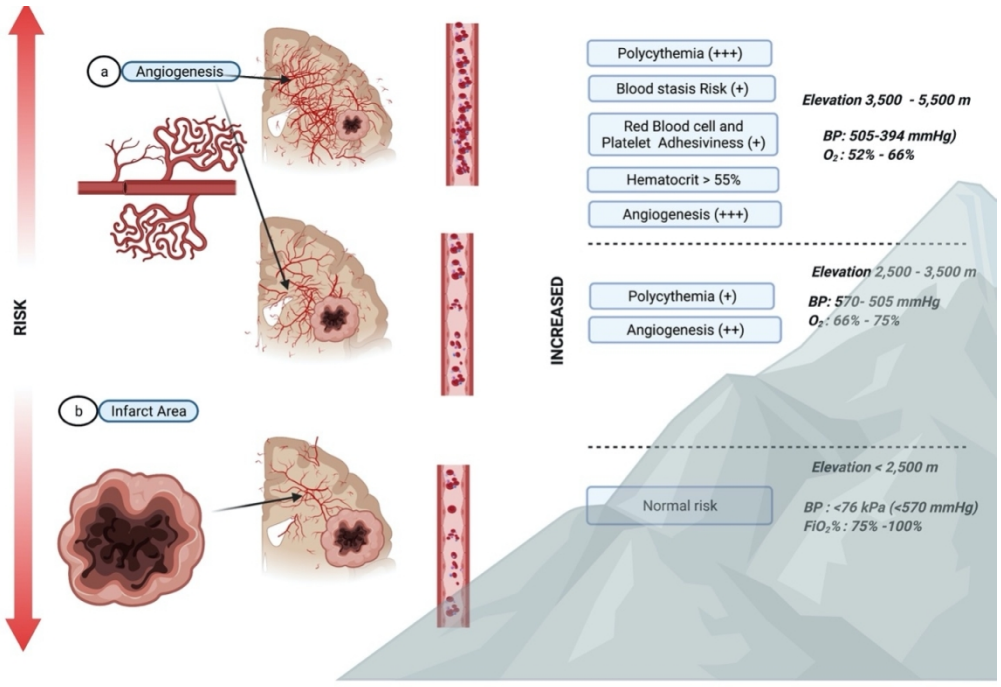


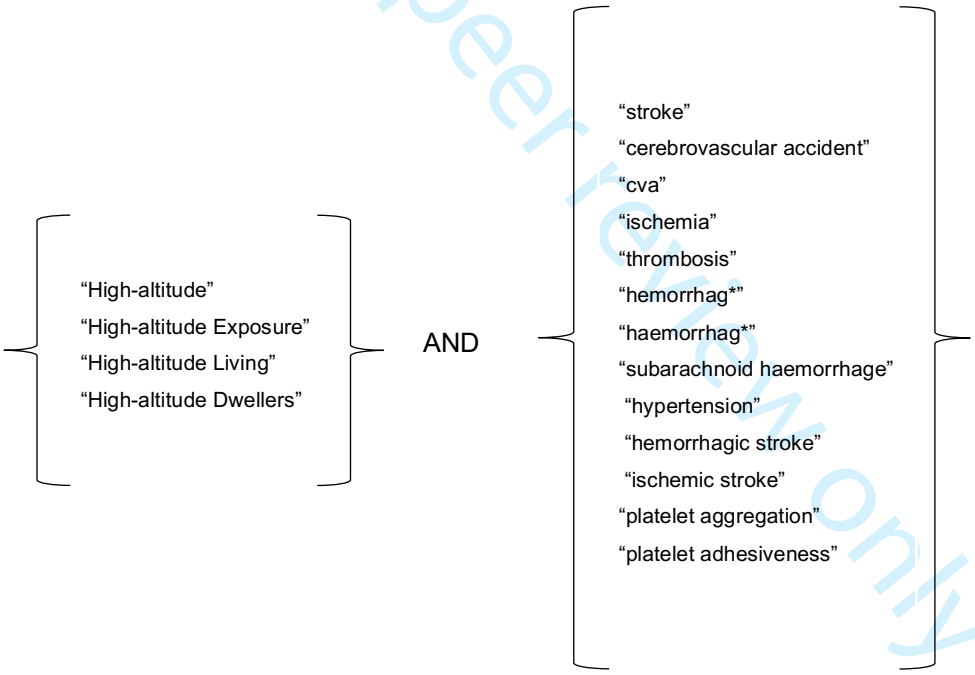
Figure 2

165x114mm (220 x 220 DPI)

Supplementary File 1

The literature review process was performed in English in order to cover the largest number of scientific databases and repositories containing academic literature from all over the world that are accessible for most of the researchers.

The following libraries, repositories and databases were accessed using the University of Southampton library tool Delphis: AMED, EMBASE, Cochrane Library, PubMed, MEDLINE and Europe PMC. The Latin-American bibliographic database Scielo was also included in a separate search from its website repository. The following terms were used during the search strategy while defined terms within the parenthesis were combined with the Boolean operator "OR".



The exclusion criteria for the literature review excluded all *in vivo* and *in vitro* studies and all the data analysed that did not concern human studies. If the search term "stroke" referred to cardiac output or stroke volume, the results were also excluded. The length of the data extraction was set from 1960 to 2021 and any study that matched the search strategy and did not counterpose the exclusion criteria were revised.

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A total of 1,890 manuscripts were retrieved during the first step of the literature review process and only those which fulfil the inclusion criteria were included for the author's revision. After this process, a total of 221 abstracts were reviewed and those that had an exclusion criterion were removed. In the end, a total of 17 manuscripts were included in this review.

List of Databases and Information providers linked to Delphis

MEDLINE Complete
MEDLINE
Complementary Index
Academic Search Complete
Business Insights Global
ScienceDirect
Gale In Context: Global Issues
Journals@OVID
PASCAL Archive
Engineering Source
Gale In Context: Middle School
vLex
Regional Business News
Applied Science & Technology Source
Gale OneFile: CPI.Q
Business Source Complete
Computers & Applied Sciences Complete
JSTOR Journals
Education Source
SocINDEX with Full Text
FRANCIS Archive
Legal Source
Humanities Source
GreenFILE
Dentistry & Oral Sciences Source
Fuente Académica Premier
Library & Information Science Source
Library, Information Science & Technology Abstracts
eBook Index
Aerospace Research Central

For peer review only

eBook Collection (EBSCOhost)

Art & Architecture Source

For peer review only

Authors	Razdanet al. 1988	Annoml et.al 1990	Garcia, J 1994	Jaillard, 1995	Jha et. al 2002	Niaz et al. 2003	Mahajan et. al 2004	Faeh et.al 2009	Ezzati et.al 2011	Dhiman et. al 2018	Khattar et.al 2019	Lu et.al 2020	Burtscher et. al 2021	Ortiz et.al 2021	Liu et.al 2021
Did the review address a clearly focused question?	YES	YES	YES	YES	YES	YES	NO	YES	NO	YES	YES	YES	YES	YES	YES
Did the authors look for the right type of papers?	CT	NO	CT	CT	YES	CT	CT	YES	CT	YES	YES	YES	YES	YES	YES
Do you think all the important, relevant studies were included?	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable
Did the review's authors do enough to assess quality of the included studies?	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable	Not appiclable
If the results of the review have been combined, was it reasonable to do so?	CT	YES	YES	NO	CT	CT	NO	CT	NO	CT	CT	CT	CT	CT	CT

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What are the overall results of the review?	YES	NO	YES	CT	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
How precise are the results?	NO	NO	NO	NO	YES	NO	NO	NO	NO	YES	YES	YES	YES	YES	YES
Can the results be applied to the local population ?	YES	NO	NO	YES	YES	YES	YES	YES	CT	YES	YES	YES	YES	YES	YES
Were all important outcomes considered ?	YES	CT	NO	YES	YES	YES	YES	NO	YES	YES	YES	YES	YES	YES	YES
Are the benefits worth the harms and costs?	CT	NO	CT	CT	YES	YES	CT	CT	CT	YES	YES	YES	YES	YES	YES

Supplementary Table 1 CASP analysis of the selected literature

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Reporting checklist for systematic review and meta-analysis.

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement

		Reporting Item	Page Number
Title			
	#1	Identify the report as a systematic review, meta-analysis, or both.	1
Abstract			
Structured summary	#2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	2
Introduction			
Rationale	#3	Describe the rationale for the review in the context of what is already known.	4

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Objectives	#4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
Methods			
Protocol and registration	#5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address) and, if available, provide registration information including the registration number.	4
Eligibility criteria	#6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rational	4
Information sources	#7	Describe all information sources in the search (e.g., databases with dates of coverage, contact with study authors to identify additional studies) and date last searched.	4
Search	#8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	#9	State the process for selecting studies (i.e., for screening, for determining eligibility, for inclusion in the systematic review, and, if applicable, for inclusion in the meta-analysis).	4
Data collection process	#10	Describe the method of data extraction from reports (e.g., piloted forms, independently by two reviewers) and any processes for obtaining and confirming data from investigators.	4
Data items	#11	List and define all variables for which data were sought (e.g., PICOS, funding sources), and any assumptions and simplifications made.	4
Risk of bias in individual studies	#12	Describe methods used for assessing risk of bias in individual studies (including specification of whether this was done at the study or outcome level, or both), and how this information is to be used in any data synthesis.	5
Summary measures	#13	State the principal summary measures (e.g., risk ratio, difference in means).	5

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Planned methods of analyis	#14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I2) for each meta-analysis.	5
Risk of bias across studies	#15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	#16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
Results			
Study selection	#17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	#18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citation.	6
Risk of bias within studies	#19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see Item 12).	5
Results of individual studies	#20	For all outcomes considered (benefits and harms), present, for each study: (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot.	7-15
Synthesis of results	#21	Present the main results of the review. If meta-analyses are done, include for each, confidence intervals and measures of consistency.	7-15
Risk of bias across studies	#22	Present results of any assessment of risk of bias across studies (see Item 15).	7-15
Additional analysis	#23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7-15
Discussion			

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Summary of Evidence	#24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., health care providers, users, and policy makers	16
Limitations	#25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	20
Conclusions	#26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	20
Funding			
Funding	#27	Describe sources of funding or other support (e.g., supply of data) for the systematic review; role of funders for the systematic review.	21

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